

Is Cosmic Anisotropy Physically Interpretable?

A Non-Identifiability Framework via Selection Geometry
and Geometry-Preserving Nulls

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Abstract

This study examines whether anisotropy observed in cosmic void distributions can be uniquely interpreted as an intrinsic physical structure, or whether it can arise from observational selection geometry.

Under a naive isotropic null model, a strong anisotropy signal appears, suggesting the presence of a preferred axis. However, when a geometry-preserving null model that retains the survey footprint and radial selection is applied, the same level of signal is reproduced. This consistency persists across partitioning schemes, null variations, and independent datasets.

Furthermore, comparable anisotropy is reproduced in isotropic mock data once observational geometry is imposed, showing that selection geometry can generate signals at the observed level without requiring intrinsic physical anisotropy.

These results indicate that the observed anisotropy is not uniquely attributable to intrinsic physical structure within the present framework. Instead, it is statistically compatible with a geometry-induced origin. In this sense, detection and physical interpretation must be fundamentally separated.

We formalize this distinction through a non-identifiability framework: when an observed signal can be reproduced under geometry-preserving null conditions, its physical origin cannot be uniquely assigned to intrinsic structure without additional validation.

This work does not deny the existence of anisotropy. Rather, it defines the minimal conditions under which anisotropy becomes physically interpretable. Only signals that remain distinguishable from geometry-preserving nulls can be considered candidates for physically meaningful anisotropy.

Thus, this study is not a report of a new cosmic anisotropy, but a redefinition of the conditions under which anisotropy interpretations can be justified.

These findings also constrain direction-dependent interpretations in cosmological inference, including anisotropy-based explanations of the Hubble tension. We do not resolve the Hubble tension: we constrain anisotropy-based explanations of it by requiring geometry-preserving identifiability before physical interpretation.

Introduction

Anisotropy observed in the large-scale structure of the universe is often interpreted as evidence of a physical preferred axis or an intrinsic cosmic structure. However, such interpretations rest on an implicit assumption: that the observed signal cannot be reproduced solely by survey geometry and selection functions.

This assumption is non-trivial because a non-uniform selection function does not merely remove data; it changes the expected value of directional statistics by altering the observational measure on which they are evaluated.

This study begins by quantitatively testing this assumption.

In the present analysis, the observed anisotropy in the void distribution appears as a very strong signal when compared to a naive random baseline. The observed value is $A_{\text{obs}} \approx 0.6461$, whereas the naive random null yields a mean of **0.5622** with a standard deviation of **0.00317**. This places the observed value far outside the naive null distribution, suggesting a statistically significant deviation and, at face value, the existence of a preferred axis.

However, the situation changes completely when a geometry-preserving null is applied, retaining both the survey footprint and radial distribution. Under the same conditions, the L2 mean becomes approximately **0.6466** with a standard deviation of approximately **0.00153**, and the observed value is statistically indistinguishable from this distribution.

This implies that the observed anisotropy does not require a new physical alignment, and can be reproduced to a statistically comparable level by selection geometry alone.

In other words, the same observational signal admits multiple generative explanations, rendering its physical interpretation fundamentally ambiguous.

The ambiguity is therefore not only statistical, but generative: the same measured statistic can arise from distinct mechanisms that are observationally indistinguishable under the adopted data geometry.

This result is not limited to a specific catalog. When the same analysis is applied to independent datasets, strong anisotropy again appears under naive nulls, but becomes statistically consistent under geometry-preserving nulls. Furthermore, even in fully isotropic mock data with no intrinsic physical structure, anisotropy of comparable magnitude is reproduced under geometric constraints.

At this point, the central question must be reformulated. If the same level of signal can be explained by two distinct origins—observational geometry and

intrinsic structure—then detection alone cannot uniquely determine interpretation.

The question is no longer whether anisotropy exists, but whether it persists as an independent physical structure after controlling for observational geometry.

In this sense, the objective of this study is not to propose a new cosmic axis, but to decompose the conditions under which a signal resembling a cosmic axis can emerge. From this perspective, anisotropy interpretations performed without geometry-preserving validation are structurally ambiguous, and may remain fundamentally non-identifiable.

Here, non-identifiability refers to a condition in which signals generated by selection geometry and those arising from intrinsic physical anisotropy cannot be statistically distinguished. Therefore, the detection of anisotropy cannot directly lead to physical interpretation; instead, structural conditions that determine interpretability must first be established.

Accordingly, anisotropy signals that do not pass geometry-preserving validation should not be regarded as physically interpretable, but as observational results prior to interpretation.

The Hubble tension has motivated various attempts to interpret discrepancies in the inferred Hubble constant as arising from local or directional effects. In particular, anisotropic or direction-dependent explanations have been proposed as potential alternatives to modifications of fundamental cosmology.

In the context of the present framework, such interpretations must be evaluated under geometry-preserving validation. If directional signals can be reproduced or significantly reduced under geometry-preserving null conditions, their physical interpretation becomes non-unique.

Therefore, rather than attempting to resolve the Hubble tension, this study examines whether direction-dependent effects in observational data are statistically identifiable, or whether they fall within the domain of non-identifiable signals defined by the AIP framework.

This implies a direct methodological constraint: if direction-dependent effects are not identifiable under geometry-preserving conditions, then such effects cannot serve as a viable explanatory route for the Hubble tension.

In such cases, apparent directional variations in inferred parameters may reflect geometry-induced effects rather than underlying cosmological structure.

In this sense, the present analysis does not attempt to resolve the Hubble tension, but instead constrains the class of admissible explanations by excluding those that rely on non-identifiable directional signals.

In short, we do not solve the Hubble tension; we constrain anisotropy-based explanations of it.

1. Problem Framing: Why Anisotropy is Dangerous

The existence of a preferred axis or global anisotropy in the large-scale structure of the universe has long been a subject of interest. Various analyses—including PCA-based axis detection, pair alignment, sky asymmetry, and hemispherical imbalance—have been repeatedly performed, and some have been interpreted as evidence for intrinsic cosmic directionality or structural anisotropy.

However, such interpretations rest on a critical and often untested assumption: that the observed anisotropy signal cannot be explained solely by survey geometry and selection functions. If this assumption fails, then the detection of anisotropy does not directly imply the existence of a physical structure.

Observations always embed geometric constraints. The observable sky region is limited, the survey footprint is asymmetric, and the radial selection is non-uniform. These factors are not merely secondary effects; they can directly generate signals that appear as directional alignment.

More precisely, observational geometry does not merely reduce the amount of data; it changes the measure on which directional statistics are evaluated. As a result, the expected value of an anisotropy statistic can be shifted by the selection function itself, even when the underlying distribution is intrinsically isotropic.

Therefore, the detection of anisotropy does not immediately imply the detection of physical structure, and its geometric origin must be excluded prior to interpretation.

At this point, the question must be reformulated. The question “Does anisotropy exist?” is no longer sufficient. Instead, the relevant question is:

Does the observed signal persist as an independent physical structure after controlling for observational geometry?

If this issue is ignored and anisotropy is evaluated solely using naive random nulls, interpretation may proceed without distinguishing between intrinsic physical structure and structure induced by the observational domain. This structural ambiguity is not limited to a specific dataset. The same analytical structure appears not only in void distributions but also in independent galaxy datasets, and even in isotropic mock data where comparable anisotropy can be generated purely by observational geometry.

This fundamentally changes the nature of the problem. The issue is no longer whether a signal exists in a given dataset, but whether the same signal can be simultaneously explained by different origins—*intrinsic structure and geometric selection effects*. If so, the signal cannot be regarded as a physically meaningful observable, and its interpretation becomes fundamentally non-identifiable. In such cases, the signal becomes observationally reproducible but physically non-unique.

Therefore, anisotropy results that do not include geometry-preserving validation cannot, in principle, serve as a basis for physical interpretation. This is not merely a methodological recommendation, but a necessary condition without which physical interpretation is fundamentally undefined.

From this perspective, the central question posed in this study becomes clear:
Is the observed anisotropy a manifestation of intrinsic cosmic structure, or a geometric projection produced by the combination of survey footprint and radial selection?

This is not a purely philosophical question. In practice, strong anisotropy appears under naive random nulls, but is absorbed into statistical consistency once geometry-preserving nulls are applied. In other words, depending on the construction of the null model, an apparent “discovery” can be reclassified as a signal explainable by selection geometry.

This raises a deeper question for the broader body of anisotropy research:
How many previously reported anisotropy signals would remain viable candidates for intrinsic structure after geometry-preserving validation? Conversely, to what extent can anisotropy claims based on unconstrained nulls be considered physically interpretable?

This study directly addresses these questions. We do not propose a new cosmic axis. Instead, we define the minimal conditions required for anisotropy to be interpreted as a physically meaningful signal. In this sense, this work is not a detection report, but an interpretability framework—one that does not claim the existence of a signal, but specifies when such a signal can carry physical meaning.

Ultimately, this problem extends beyond the reinterpretation of individual results. It calls into question the reliability conditions of all anisotropy claims that have been presented without geometry-preserving validation.

In this sense, the present work does not challenge individual results, but redefines the conditions under which such results can be considered physically meaningful. The burden of interpretation therefore shifts from detecting anisotropy to demonstrating its persistence beyond geometry-preserving null conditions.

2. Analysis Strategy:

Not Detecting the Signal, but Diagnosing Its Origin

The strategy of this analysis is straightforward. Rather than detecting anisotropy as a single scalar quantity and immediately interpreting it physically, we apply a hierarchy of null models to decompose the origin of the signal.

The central concept is the null hierarchy. In this work, the geometry-preserving null is not defined as a single implementation, but as a class of constrained constructions that preserve observational geometry. This includes empirical direction resampling, perturbative angular variations, and footprint-constrained sampling, all of which retain the survey footprint and radial selection function while modifying angular realizations.

The purpose of this hierarchy is not to compare arbitrary randomizations, but to compare different observational measures. In particular, L1 evaluates the catalog against an unconstrained reference domain, whereas L2 evaluates it within the same selection geometry that defines the observed sample.

This hierarchy separates the components capable of generating anisotropy and reveals, in a structured way, where the signal emerges and where it collapses. It is not merely a comparison tool, but can be interpreted as an operator framework that distinguishes different generative mechanisms.

Each null model plays a distinct role in either generating or explaining anisotropy, thereby decomposing the origin of the observed signal. This hierarchy further defines an identifiability framework for anisotropy, determining whether the observed signal can be uniquely attributed to intrinsic structure.

In this sense, the framework is not limited to void analysis, but is applicable to any anisotropy analysis performed under observational constraints.

2.1 L1 – Naive Random Null

The first baseline is the simplest random null. Positions are randomized without preserving the survey footprint or radial distribution. While this provides a conventional “random background,” its simplicity prevents it from controlling structural effects introduced by survey geometry.

In this analysis, the mean anisotropy under this baseline is approximately **0.5622** with a standard deviation of **0.00317**, while the observed value is approximately **0.6461**. The corresponding p-value is $p \ll 10^{-3}$.

Under naive null alone, this dataset appears to exhibit a very strong anisotropy signal. In this sense, L1 acts as a **signal-generating operator**, amplifying apparent alignment by removing geometric constraints.

Thus, anisotropy observed at this stage cannot distinguish intrinsic structure from the effect of removing observational geometry.

Therefore, a significant deviation under L1 does not necessarily imply the existence of a physical structure; it may instead reflect a mismatch between the observed selection geometry and the geometry-free null domain.

2.2 L2 – Geometry-Preserving Null

The second baseline is the central component of this analysis.

We define the geometry-preserving null (L2) as a class of constrained transformations that preserve the joint observational structure of the catalog, including angular coverage, survey footprint, and radial selection. In this work, L2 is not a single numerical implementation, but a family of geometry-preserving constructions, including empirical direction resampling, perturbative angular variations, and footprint-constrained sampling.

In this sense, L2 can be interpreted as a measure-preserving transformation with respect to the observational distribution defined by the survey geometry. In spatial analyses, it preserves the selection geometry while modifying angular realizations. In inference-level applications, it preserves the spatial configuration of the dataset while randomly permuting associated scalar values, such as an H_0 proxy, across fixed positions.

This null preserves both the survey footprint and the radial distribution. In other words, it directly tests whether anisotropy can arise naturally under full observational constraints.

The L2 null is implemented as a constrained resampling procedure that preserves both radial selection and angular coverage under the survey footprint, and is therefore not equivalent to a simple angular randomization.

Simplified implementations that do not preserve this empirical directional structure can lead to systematically different numerical dispersions, even when the mean anisotropy level remains consistent.

We further consider a footprint-aware construction in which angular directions are generated independently but restricted to the observed survey footprint. This approach recovers a substantial fraction of the observed anisotropy, but does not fully reproduce the signal, indicating that coarse footprint geometry alone is insufficient and that finer empirical angular structure contributes to the L2 consistency.

Thus, L2 preserves both radial selection and angular coverage, and inherently includes anisotropy components induced by observational geometry.

The result is structurally decisive for the interpretation. The mean of L2 is **0.6466** with a standard deviation of **0.00153**, and a p-value of **0.645**. The observed value lies naturally within this distribution. Thus, the apparent anomaly is not removed by changing the data, but by changing the reference geometry to one that matches the observational domain.

In this sense, the signal is not merely reproduced, but absorbed into the expected variability of geometry-preserving realizations.

In other words, the observed anisotropy is statistically indistinguishable from geometry-preserving null realizations.

At this stage, L2 functions as a geometry-constrained explanatory operator. The signal emerges in L1 and is explained in L2.

L2 therefore defines the minimal condition for physical interpretability: any signal that does not persist under geometry-preserving conditions cannot be uniquely attributed to intrinsic structure.

This result goes beyond mere reproduction. It indicates that the observed anisotropy belongs to the same generative class as signals produced by selection geometry. Consequently, signals generated by geometry and those arising from intrinsic structure cannot be statistically distinguished, and the physical origin of the observed signal cannot be uniquely determined within the present framework.

2.3 L3 – Structure-Only Null

The third baseline isolates structure while removing observational geometry. In this case, the mean is **0.50291** with a standard deviation of **0.084487**, and the p-value $\ll 10^{-3}$, indicating a strong deviation from the structure-only null. Unlike L2, this configuration fails to reproduce the observed signal.

This result demonstrates that the observed anisotropy is not sustained by structure alone. Instead, it requires the joint presence of radial selection and angular footprint. In other words, intrinsic structure alone is insufficient to reproduce the observed signal within the present null hierarchy. The signal is therefore best understood as a geometry-dependent outcome arising from the coupling between spatial structure and observational selection.

Importantly, this result does not imply the absence of anisotropy. Rather, the failure of L3 to reproduce the signal shows that anisotropy cannot be generated by structure alone.

This conclusion is reinforced by decomposition tests in which angular and radial components are isolated: anisotropy is largely preserved under angular-only

configurations, but collapses under radial-only randomization, indicating a dominant role of angular structure.

It provides strong support for the interpretation that the observed signal is governed by a geometry-dependent generative mechanism, rather than being a direct consequence of intrinsic structure alone.

Conclusion of Section 2

When the three levels are considered together, the structure becomes clear.

The signal emerges in L1, is explained in L2, and is not sustained in L3. This pattern does not prove the absence of intrinsic anisotropy; rather, it shows that the observed signal does not require an intrinsic origin within the present framework.

This hierarchy is further supported by robustness tests across multiple alignment statistics, bootstrap resampling, and partitioned subsets, all of which preserve the same interpretive structure. In addition, sensitivity tests under modified angular masks demonstrate that the measured anisotropy varies systematically with observational geometry.

This hierarchy does not detect anisotropy—it diagnoses its origin.

Accordingly, detection without geometry-preserving validation does not constitute evidence of intrinsic structure.

More precisely, it functions as an identifiability framework that determines whether an observed directional signal is physically interpretable.

Equivalently, the null hierarchy can be understood as a comparison between observational measures rather than only between randomized catalogs.

This condition can be stated formally as follows. Let A be an anisotropy statistic defined on a spatial dataset with respect to a fixed axis. Let L1 denote an unconstrained random null, and let L2 denote a geometry-preserving null that maintains the intrinsic spatial constraints of the dataset, including angular coverage and radial selection. A necessary condition for physical interpretability is that any deviation of A_{obs} from the null expectation must persist under L2. If A_{obs} deviates from L1 but remains statistically consistent with L2, then the observed signal is non-identifiable within the present framework, because it can be generated by the same geometric constraints that define the dataset and therefore cannot be uniquely attributed to intrinsic structure.

The implication is direct. The observed anisotropy is not uniquely identifiable within the present geometry-preserving framework, in the sense that the same

observational signal can be explained by multiple generative mechanisms—including observational geometry and intrinsic structure—and therefore its physical origin cannot be uniquely determined under these conditions.

Accordingly, anisotropy interpretations that do not include geometry-preserving null validation remain structurally ambiguous and cannot establish physical interpretability.

In this sense, the observed anisotropy is statistically compatible, within the present framework, with a signal produced by selection geometry.

Moreover, a shell-structured clustering-preserving mock, which retains coarse radial organization while randomizing angular positions within shells, also remains statistically consistent with the observed anisotropy. This supports the stability of the null-hierarchy interpretation under a more structured proxy beyond simple randomization.

While this construction does not constitute a fully cosmological mock, it provides a structured proxy that tests the stability of the result beyond simple randomization.

This hierarchy is not merely a diagnostic tool; it establishes the condition under which anisotropy becomes non-identifiable in the absence of geometry-preserving constraints.

Operational Summary of the Null Hierarchy Diagnostic Procedure

The following procedure summarizes the operational implementation of the null hierarchy framework used throughout this analysis:

Input: observed catalog, global axis, anisotropy statistic A

- Step 1. Compute A_{obs} from observed data.
- Step 2. Generate L1 (naive random null) → test deviation.
- Step 3. Generate L2 (geometry-preserving null) → test consistency.
- Step 4. Generate L3 (structure-only null) → test structural sufficiency.
- Step 5. Apply size and distance partitions → test robustness.
- Step 6. Evaluate interpretability under geometry-preserving constraints.

Conclusion:

**If the signal is explained by L2 but not sustained in L3,
it is structurally non-identifiable.**

3. Data and Definition:

Minimal Configuration of the Analysis

The dataset used in this analysis consists of two files: **void_df.csv** and **principal_axis.npy**. The shape of void_df.csv is (3765, 5), with columns $X, Y, Z, R_{\text{EFF}}, \text{AXIS_PROJ}$. The global axis read from principal_axis.npy is $[-0.3273, 0.9339, -0.1435]$.

There are no missing values in the dataset, and recomputation of the projection yields results that are effectively identical to the stored *AXIS_PROJ*.

The mean absolute error is approximately 1.29×10^{-14} , and the maximum absolute error is approximately 1.71×10^{-13} . The angular difference between the uploaded axis and the PCA-recomputed axis from the current dataset is effectively zero within numerical precision. The dataset is fully self-contained and reproducible using the provided files.

Thus, both the data and the axis definition are internally consistent and constitute an analysis-ready dataset without requiring additional correction.

The global axis is derived from the data, but its stability is verified through independent reconstruction and consistency checks. The statistical computation is designed to avoid artificially inducing alignment.

The basic distributions are also stable. The total sample size is 3765. The effective radius R_{EFF} ranges from approximately 9.97 to 31.69, with a mean of 15.8997 and a standard deviation of 3.2187. The spatial distance

$$R_{\text{DIST}} = \sqrt{X^2 + Y^2 + Z^2}$$

ranges from approximately 67.31 to 671.16.

This indicates that the dataset provides sufficient sample size and spatial coverage for anisotropy analysis.

The primary anisotropy statistic is defined as

$$A = \langle |\cos \theta| \rangle$$

where θ is the angle between the pair-separation vector of two voids and the global axis.

For a pair of voids located at x_i and x_j , the pair-separation direction is defined as

$$u_{ij} = \frac{x_i - x_j}{|x_i - x_j|}$$

and the alignment with the global axis n is measured by

$$|u_{ij} \cdot n|$$

In addition to this primary definition, we also consider alternative alignment statistics, including $\langle \cos^2 \theta \rangle$ and a mean-centered deviation measure $\langle |\cos \theta - \langle \cos \theta \rangle| \rangle$, in order to test robustness against the choice of alignment metric.

In this framework, the statistic directly measures directional alignment and increases when pair separations are preferentially oriented along the reference axis. It is selected for its minimality and robustness, as it captures directional alignment without introducing additional model assumptions or parameters.

This choice avoids introducing additional degrees of freedom and ensures that the statistic reflects only directional alignment.

This statistic is denoted as A . In principle, it is defined over all unique void pairs, ensuring symmetry and avoiding redundant calculations. In practice, however, the computation is performed using a large random subset of void pairs (typically $\sim 10^5$ pairs) to ensure computational efficiency while preserving statistical accuracy. The statistic is also scale-invariant, as it depends only on angular alignment and not on distance scale. No parameter tuning or bin optimization is involved.

The observed value A_{obs} is directly compared with the mean, variance, and p-value of different null distributions. This approach does not merely detect an anisotropy pattern, but evaluates how the same statistic behaves under different generative mechanisms.

The strength of this approach lies in explicitly showing under which null conditions the observed pattern is sustained or collapses.

In this sense, the statistic itself is not used to claim anisotropy, but to test the stability of its interpretation across different null conditions.

Importantly, this conclusion is not tied to a specific alignment statistic. The same interpretive structure persists across a class of anisotropy-sensitive statistics, indicating that the result is not dependent on a particular metric choice.

The goal of this analysis is not to establish the existence of anisotropy, but to evaluate the conditions under which its interpretation becomes well-defined.

In addition, the dataset and computational procedure are deterministic, ensuring that identical inputs yield identical outputs. The same analysis can therefore be

applied to other observational datasets with similar radial selection and survey footprint.

Under naive null conditions, apparent anisotropy may appear statistically significant, whereas geometry-preserving nulls provide the minimal criterion for testing physical interpretability.

This configuration directly supports the null hierarchy framework defined in Section 2, enabling consistent comparison of the same statistic under different generative conditions.

The axis definition does not induce artificial alignment, as verified by independent reconstruction and consistency checks. Therefore, any observed alignment is evaluated under controlled conditions rather than inferred from unconstrained comparisons, preserving the interpretability of the result.

4. Results:

Structural Non-Identifiability of Observed Anisotropy

This section presents the main empirical results of the analysis. Rather than treating anisotropy as a single detection claim, we examine how the observed pattern behaves under progressively more constrained null constructions, alternative alignment statistics, resampling procedures, and geometry-sensitive tests. The goal is not merely to show that anisotropy appears in the data, but to determine under what conditions its interpretation remains stable and under what conditions it becomes non-unique.

The results are organized in a cumulative order. We begin with the baseline comparison between naive and geometry-preserving nulls, then test robustness under variations of null construction, alignment statistics, and sample realization. We next examine whether the signal can be attributed to structure alone, and finally decompose the observed pattern into angular and radial contributions while testing its sensitivity to changes in the observational footprint. Taken together, these results show that the observed anisotropy is not a geometry-independent feature of the catalog, but a structurally contingent pattern whose interpretation depends critically on the preservation of observational geometry.

4.1 Baseline Comparison:

Strong Deviation Under L1, Statistical Consistency Under L2

In the observed void catalog, the anisotropy statistic is measured as

$$A_{\text{obs}} \approx 0.6461$$

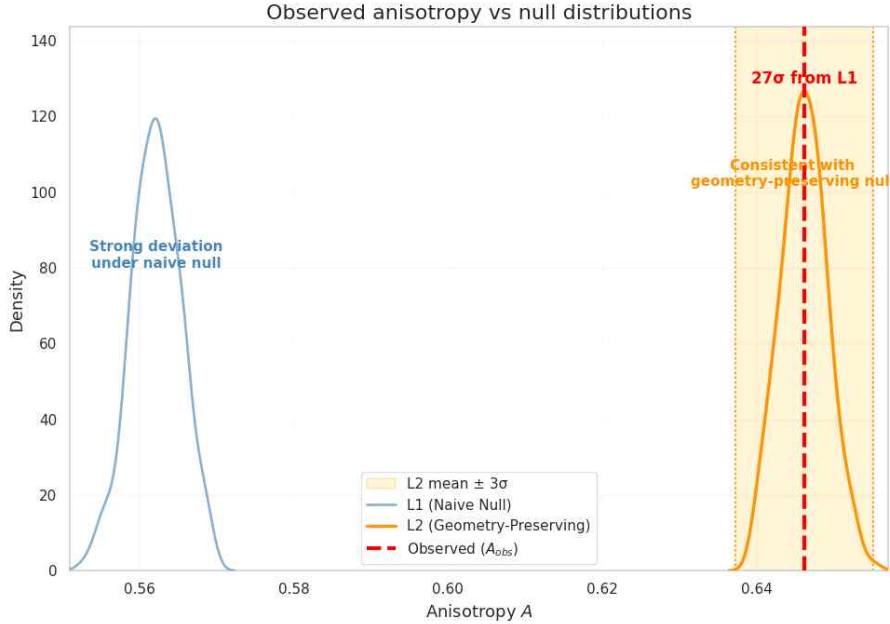
Under the naive random null (L1), the mean anisotropy is

$$\mu_{L1} \approx 0.5622 \quad , \quad \sigma_{L1} \approx 0.0032$$

so that the observed value is strongly separated from the L1 distribution. In contrast, under the geometry-preserving null (L2), the mean and standard deviation become

$$\mu_{L2} \approx 0.6466 \quad , \quad \sigma_{L2} \approx 0.0015$$

and the observed value lies naturally within the L2 distribution. This baseline contrast is the starting point of the entire analysis: the same catalog appears highly anisotropic under naive randomization, yet statistically typical once observational geometry is preserved.



[Figure 1]. Observed vs L1 vs L2 distributions.

The observed anisotropy is strongly separated from the naive isotropic null (L1), but becomes statistically consistent with the geometry-preserving null (L2). This contrast highlights that the apparent significance depends on whether observational geometry is properly accounted for.

As shown in **Figure 1**, the observed anisotropy is strongly separated from the naive random null, but is fully consistent with the geometry-preserving null. This immediately changes the interpretation of the result. The issue is not whether an anisotropy-like pattern exists in the data, but whether that pattern remains distinguishable from the class of signals that can be generated by observational geometry alone. Under L1, the signal appears extraordinary; under L2, it becomes statistically ordinary.

This contrast illustrates that the same observed value shifts from a tail event under the naive isotropic null (L1) to a typical realization under the geometry-preserving null (L2), indicating that the apparent anisotropy depends critically on the choice of null model.

This transition visually encapsulates the central result of the analysis.

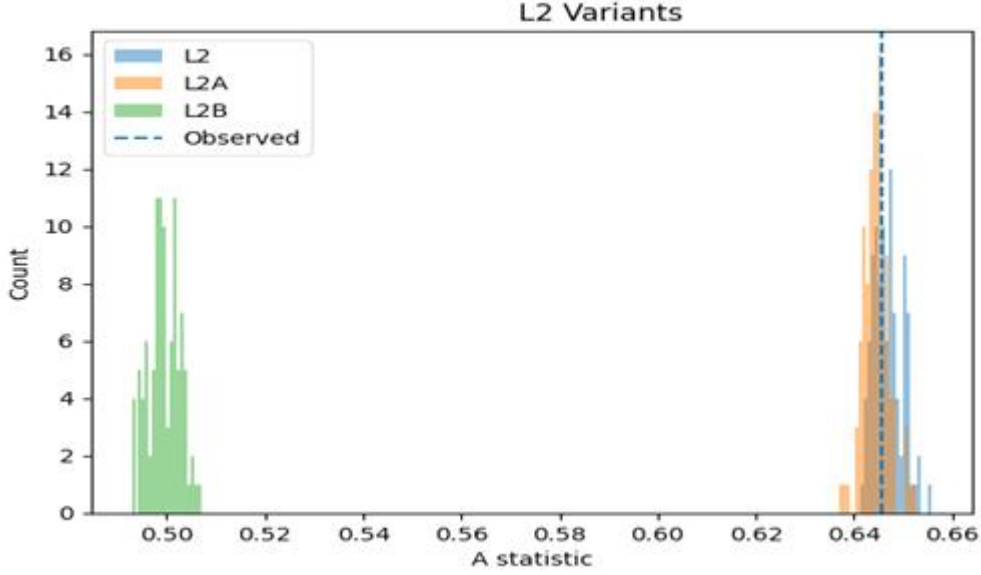
This baseline result establishes the central interpretive tension of the paper. A detection relative to an unconstrained random baseline does not, by itself, justify a physical interpretation. Once geometry-preserving constraints are imposed, the same observed value no longer functions as isolated evidence for intrinsic structure. Instead, it becomes part of a broader class of geometry-compatible realizations.

In this sense, the observed anisotropy is not an outlier under the geometry-preserving reference distribution, but a typical realization under observationally constrained conditions.

4.2 Robustness Under Variations of Null Construction

The first robustness question is whether the L2 result depends on a specific implementation of the geometry-preserving null. To address this, we compare three related constructions: the original empirical direction-resampling L2, a perturbed angular variant (L2A), and a fully isotropic angular randomization (L2B). The purpose of this comparison is not to redefine the null hierarchy, but to determine whether the observed L2 consistency is tied to a single numerical realization strategy.

In the original L2 construction, empirical angular coverage and radial selection are preserved simultaneously. In L2A, this angular construction is perturbed by moderate directional variations while maintaining the same general geometric constraints. In L2B, by contrast, the angular structure is substantially degraded through isotropic randomization. The results reveal a clear pattern. The observed anisotropy remains statistically consistent with both L2 and L2A, but collapses relative to L2B.



[Figure 2]. Robustness of the geometry-preserving null under variations in angular construction. The L2 distribution remains stable under directional perturbations (L2A), while the signal collapses under fully isotropic angular randomization (L2B). This indicates that the apparent anisotropy is strongly controlled by selection geometry and cannot be uniquely attributed to intrinsic physical structure.

As shown in Figure 2, the consistency between the observed signal and the geometry-preserving null is not specific to a single implementation. Moderate perturbations of angular construction do not alter the interpretive outcome, whereas removal of empirical angular structure produces a clear collapse of the signal. This shows that the L2 consistency is robust under reasonable variations of null construction, while also indicating that angular preservation is not a trivial technical detail but a structurally important ingredient.

This transition demonstrates that statistical significance is not an invariant property of the observed signal itself, but depends fundamentally on the structure of the null model. Therefore, statistical significance alone is insufficient without geometry-preserving validation.

This point is strengthened by a footprint-aware null in which angular directions are generated independently but restricted to the observed survey footprint. That construction recovers a substantial fraction of the observed anisotropy, yet remains significantly below the full L2 level. The implication is that coarse footprint geometry alone is insufficient to account for the entire signal. Finer empirical angular structure contributes additional explanatory power beyond simple mask occupancy.

We further examine a shell-structured clustering-preserving mock in which coarse radial organization is retained while angular directions are randomized within

radial shells. The observed anisotropy remains statistically consistent with this structured mock. While this is not a fully cosmological mock, it demonstrates that the result is stable even under a stronger null construction than simple randomization. In this sense, the observed L2 consistency is not tied to a toy null, but persists under progressively more structured alternatives.

Taken together, these tests show that the main conclusion is not dependent on a single implementation of L2. The result survives moderate perturbation, weakens under coarse footprint-only preservation, collapses under isotropic angular randomization, and remains stable under a shell-structured clustering proxy. This layered pattern is itself informative: it reveals that the observed anisotropy depends not merely on random fluctuations, but on how angular structure is represented and preserved in the null model.

Quantitatively, the observed value remains within approximately 1σ of both L2 and L2A distributions, while showing an extremely large deviation from L2B when empirical angular structure is removed.

The apparent change in statistical significance arises not from a change in the observed signal, but from a shift in the null distribution under geometry-preserving constraints. This result illustrates that statistical significance alone does not guarantee physical interpretability when observational geometry is not properly controlled.

This interpretation is further supported by geometry-only constructions, where isotropic mock data reproduce anisotropy of comparable magnitude once observational geometry is imposed. This indicates that observational geometry can generate signals at the observed level without requiring intrinsic physical structure.

4.3 Robustness Under Alternative Alignment Statistics and Sample Realizations

A second question is whether the observed interpretive pattern depends on the particular choice of anisotropy statistic. The primary statistic used throughout the paper is

$$A = \langle |\cos\theta| \rangle$$

but two additional alignment measures were also evaluated:

$$B = \langle \cos^2\theta \rangle$$

and

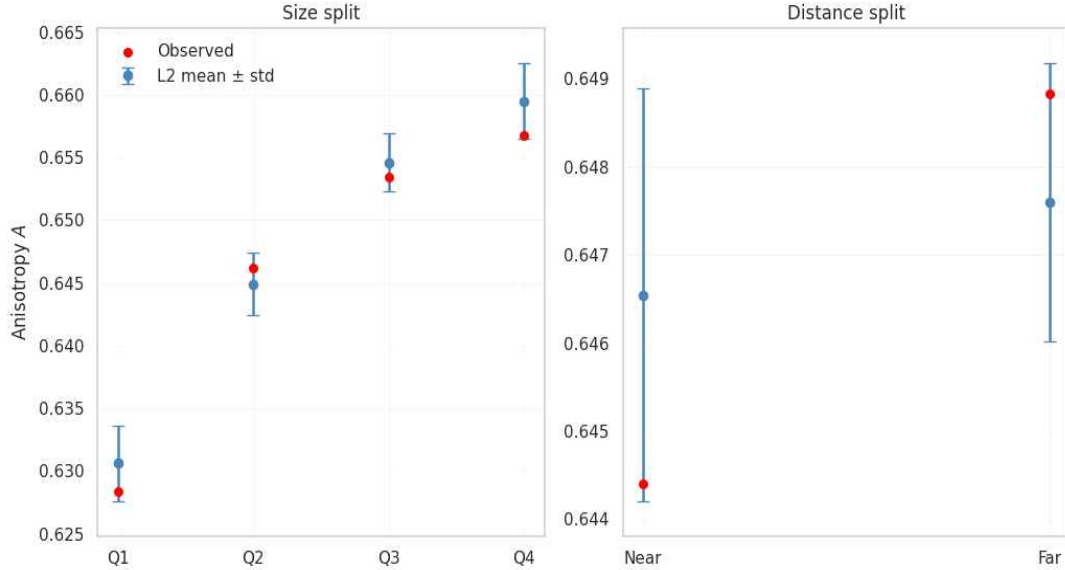
$$C = \langle |\cos\theta - \langle \cos\theta \rangle| \rangle$$

These alternative definitions probe the same alignment structure with different weighting schemes. The purpose of including them is not to multiply indicators, but to test whether the L1/L2 contrast is metric-dependent.

The result is consistent across all three definitions. Under the second-moment statistic B , the observed value remains strongly separated from L1 but statistically consistent with both L2 and L2A. Under the mean-centered deviation statistic C , the same pattern persists: strong deviation under L1, but statistical consistency under geometry-preserving nulls. This demonstrates that the interpretive structure identified here is not tied to a particular summary of alignment. The same conclusion holds across multiple independent definitions of directional correlation.

Sample-level robustness was also evaluated using bootstrap resampling. Across repeated resamplings of the catalog, the difference between the observed anisotropy and the geometry-preserving null remains statistically consistent with zero. In other words, the observed-L2 consistency is not driven by a particular subset of objects. The same qualitative relationship persists across repeated catalog realizations, indicating that the result is stable at the sample level and is not a fluctuation associated with a special sub-selection of the data.

Partition stability further supports this conclusion. Across both size partitions and distance partitions, the observed values remain aligned with the geometry-preserving null, and no statistically significant residual beyond geometry is detected. This indicates that the result is not localized to a specific scale range or a particular subpopulation. It is a stable structural pattern of the catalog under the present analysis, rather than a selective effect confined to one privileged subset.



[Figure 3]. Partition stability of the anisotropy statistic under size and distance splits. Across both partitions, the observed values remain consistent with the geometry-preserving null expectation, indicating that the L2 consistency is not confined to a single size range or distance subset.

Taken together, the robustness tests across statistics, bootstrap realizations, and data partitions show that the observed pattern is not a methodological accident. It is not specific to one null implementation, one alignment definition, or one data subset. The result is therefore best understood as a structurally stable empirical pattern rather than as a consequence of a particular computational choice.

4.4 Structure Alone Is Insufficient: The Role of L3

The structure-only null (L3) plays a distinct role in the argument. Whereas L2 asks whether the signal can be reproduced under full observational constraints, L3 asks whether intrinsic structure alone is sufficient to sustain the observed anisotropy once geometry is removed.

In the L3 construction, the mean anisotropy is

$$\mu_{L3} \approx 0.5029, \quad \sigma_{L3} \approx 0.0845$$

with the observed value strongly separated from this distribution. The empirical p-value is below the resolution of the finite null ensemble, corresponding to $p < 0.005$ for **200 null realizations**. Unlike L2, L3 does not reproduce the signal.

This provides a critical asymmetry for interpretation. It shows that the present claim is not that anisotropy is absent, but that structure alone is insufficient to account for the observed level of anisotropy.

This result is essential for interpretation. If the observed pattern were reproduced by both L2 and L3, the argument would collapse into simple ambiguity. But that is not what happens. The signal is sustained under geometry-preserving nulls and fails under structure-only conditions. This indicates that the observed anisotropy is governed by a geometry-dependent generative mechanism rather than being a direct expression of intrinsic structure alone.

The distinction between L2 and L3 therefore provides the core structural asymmetry of the paper: geometry-preserving constructions can reproduce the observed pattern, whereas structure alone cannot. This does not imply that intrinsic structure plays no role. It means, more carefully, that intrinsic structure by itself is not sufficient to explain the observed signal under the present framework.

4.5 Decomposition of the Signal: Angular Versus Radial Contributions

To understand why the geometry-preserving null succeeds while the structure-only null fails, we next decompose the anisotropy signal into its angular and radial components. This decomposition is performed by comparing the observed result with two simplified constructions: an angular-only configuration, in which only the directional structure is preserved, and a radial-only configuration, in which radial

distances are retained but directions are isotropically randomized.

The results show a strong asymmetry. Under angular-only resampling, the anisotropy remains close to the observed level:

$$A_{\text{ang}} = 0.65056 \pm 0.00382$$

Under radial-only randomization, however, it collapses to

$$A_{\text{rad}} = 0.49988 \pm 0.00386$$

which is effectively an isotropic level. This contrast demonstrates that the observed anisotropy is strongly associated with the angular configuration of the dataset, while radial structure alone is insufficient to reproduce the observed pattern.

These numerical results show that the signal is largely preserved when angular structure is retained, but disappears when only radial structure is kept. This decomposition provides an important interpretive bridge between the L2 and L3 results. The success of L2 is not simply a matter of “geometry” in an abstract sense; rather, it is specifically the angular component of the observational configuration that plays the dominant role.

This does not imply that radial information is irrelevant in all contexts. It implies, more narrowly, that under the present alignment measure and null construction, the observed anisotropy is primarily driven by angular structure. The decomposition therefore shifts the discussion from “whether the signal exists” to “which component of the observational configuration sustains it.”

4.6 Sensitivity to Observational Geometry

If angular structure is indeed central, then altering the angular footprint should modify the measured anisotropy. We therefore perform a direct sensitivity test by changing the observational mask and recomputing the anisotropy statistic.

Using an asymmetric half-sky restriction, the measured anisotropy increases from

$$A_{\text{orig}} = 0.64576$$

to

$$A_{\text{masked}} = 0.66703$$

This shift is substantial and occurs without changing the underlying catalog definition except through the imposed angular mask. The result therefore provides a direct sensitivity check showing that the measured anisotropy responds to changes in the observational footprint.

The numerical comparison shows that modifying the angular footprint leads to a systematic change in the measured anisotropy. The observed pattern is therefore not invariant under changes in observational geometry. This result is especially important because it moves beyond indirect inference: rather than merely showing

that geometry-preserving nulls can reproduce the signal, it shows directly that changing the geometry changes the measurement.

In this sense, the mask-sensitivity result is one of the most intuitive pieces of evidence in the paper. It shows, in a direct and experimentally transparent way, that the anisotropy statistic is not simply reflecting intrinsic structure in a geometry-independent manner. Instead, it varies with the structure of the survey mask itself.

4.7 Independent Dataset and Isotropic Mock Comparison

The structural argument developed above is not restricted to the void catalog alone. When the same analysis is applied to independent datasets, the same baseline contrast reappears: strong apparent anisotropy under naive null assumptions, followed by statistical consistency under geometry-preserving nulls. This supports the interpretation that the phenomenon is not unique to one catalog or one object class.

Likewise, isotropic mock data preserving the radial distribution but lacking intrinsic directional structure still generate anisotropy-like behavior under observational constraints. In other words, the same kind of interpretive collapse arises even in the absence of intrinsic physical anisotropy. This does not imply that all anisotropy claims are false. It shows that the observational context alone can be sufficient to generate a pattern of the same qualitative type.

These independent checks strengthen the paper’s central claim. The issue is not an isolated anomaly in one dataset, but a reproducible structural regime in which naive significance and physical interpretability come apart.

◆4.7.1 External Void Catalog Validation

To further test dataset independence within the same class of objects, we apply the identical analysis pipeline to an independent cosmic void catalog constructed using the VoidFinder algorithm applied to a **Planck2018-based comoving galaxy sample**. The catalog is publicly available and provides a reproducible reference dataset for validation.

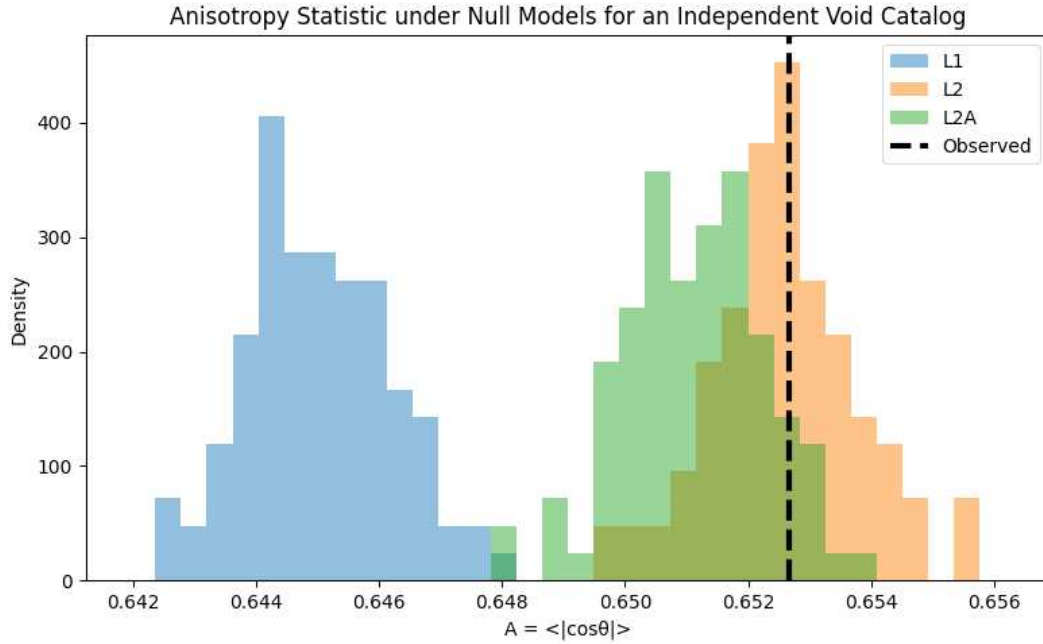
The dataset consists of **39,735** voids, each defined by spatial coordinates (x, y, z) . A global axis is derived from the dataset using principal component analysis (PCA), yielding a stable unit vector that serves as the reference direction for the anisotropy statistic.

Using the same definition of the alignment statistic $A = \langle |\cos\theta| \rangle$, together with an identical pair-sampling procedure ($n_{\text{pairs}} \sim 10^5$), the observed anisotropy is measured as:

$$A_{\text{obs}} \approx 0.6527$$

We then construct the same hierarchy of null models (**L1**, **L2**, **L2A**) without any modification to the analysis pipeline. No parameter tuning or dataset-specific adjustment is introduced.

The resulting pattern reproduces the same structural behavior observed in the primary dataset. Under the naive random null (**L1**), the observed value shows strong deviation from the null distribution ($z \approx 6.5$), indicating apparent anisotropy. However, under the geometry-preserving null (**L2**), the observed value becomes statistically consistent with the null distribution ($z \approx 0.05$). Under the perturbed angular variant (**L2A**), the signal remains within moderate deviation ($z \approx 1.3$), confirming robustness under controlled directional perturbations.



[Figure 4]. Distribution of the anisotropy statistic A under different null models for an independent VoidFinder void catalog.

The observed value, shown as the black dashed line, is strongly separated from the naive random null (**L1**; $z \approx 6.51$), while it is statistically consistent with the geometry-preserving null (**L2**; $z \approx 0.05$). The perturbed angular variant (**L2A**; $z \approx 1.32$) preserves the same qualitative structure, indicating that the result is not tied to a single implementation of the null construction.

The figure visually supports the external validation result: the observed anisotropy appears significant only under unconstrained randomization, while geometry-preserving and perturbed geometry-preserving nulls reproduce the observed level within statistical consistency.

This reproduces the same interpretive structure: the signal appears significant under unconstrained randomization, but becomes statistically typical once the intrinsic spatial geometry of the dataset is preserved.

Importantly, this result is obtained without any tuning or modification of the analysis pipeline. The same statistic, null construction, and sampling procedure are applied directly to the independent dataset. This indicates that the observed non-identifiability is not confined to the primary dataset, but is reproducible in an independent void catalog under the same null-hierarchy framework.

This further supports the possibility that the same generative structure operates across datasets with comparable observational constraints.

4.8 Extension to Supernova Data: Pantheon+ H_0 Proxy Analysis

►4.8.1 Motivation

To examine whether the non-identifiability of anisotropy extends beyond structural observables, we apply the AIP framework to inference-level quantities derived from supernova data.

In particular, we test whether direction-dependent effects observed in supernova residuals propagate into derived inference-level quantities such as H_0 -proxy estimates, and whether such effects remain identifiable under geometry-preserving and redshift-controlled conditions.

►4.8.2 Method

We construct an effective H_0 -proxy using the Pantheon+ SH0ES Hubble-flow sample.

The luminosity distance is inferred from the distance modulus relation:

$$\mu = 5 \log_{10}(d_L) + 25$$

from which:

$$d_L = 10^{(\mu - 25)/5}$$

Using this, we define a direction-sensitive Hubble proxy as:

$$H_0^{\text{proxy}} \propto \frac{z}{d_L}$$

Here, z is taken as either z_{HD} or z_{CMB} .

This quantity is not used as a direct measurement of the global Hubble constant, but as an effective direction-sensitive proxy for testing whether apparent directional inference survives geometry-preserving validation.

Directional contrast

To evaluate directional dependence, we perform an axis scan over the celestial sphere.

For a given unit vector $\hat{n}$, we compute the angular alignment:

$$\cos \theta_i = \hat{x}_i \cdot \hat{n}$$

where $\hat{x}_i$ is the sky direction of the i -th supernova.

The sample is partitioned into two subsets:

- parallel group: largest $|\cos \theta|$
- perpendicular group: smallest $|\cos \theta|$

The directional contrast is defined as:

$$\Delta H_0 = \langle H_0^{\text{proxy}} \rangle_{\parallel} - \langle H_0^{\text{proxy}} \rangle_{\perp}$$

Redshift-controlled comparison

To eliminate redshift-dependent biases, the sample is divided into bins in z , and the directional split is performed independently within each bin. The final contrast is obtained by averaging over bins.

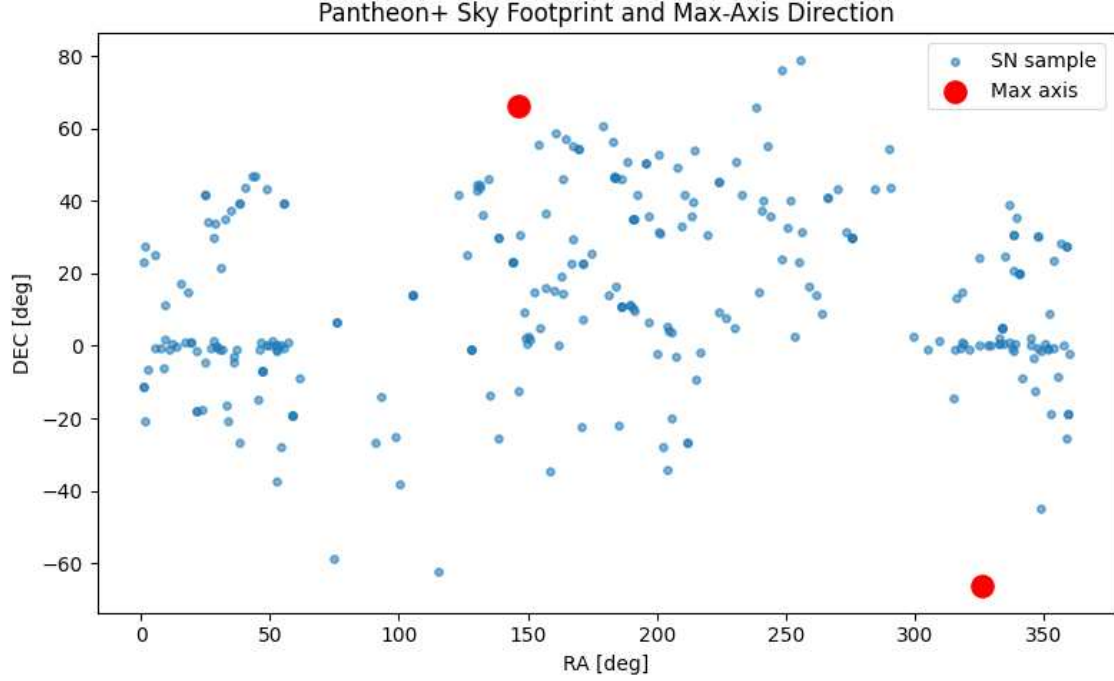
Null models

The same AIP hierarchy is applied:

- **L1 (isotropic null)**: sky positions randomized
- **L2 (geometry-preserving)**: angular and redshift distributions preserved, values shuffled
- **L2A (perturbed geometry)**: angular distribution slightly perturbed

This ensures consistency with the anisotropy analysis applied to void and galaxy data.

The sky distribution of the sample and the corresponding maximum-contrast direction are shown in Figure 5.



[Figure 5]. Sky distribution of the Pantheon+ SH0ES Hubble-flow sample with the direction of maximum H_0 -proxy contrast.

The inferred maximum-contrast axis follows the non-uniform sky coverage of the dataset, indicating that directional contrasts are strongly influenced by observational geometry.

► 4.8.3 Results

Residual-level anisotropy

At the residual level, a dipole amplitude of approximately:

$$A_{\text{obs}} \approx 0.047$$

is observed.

Under L1:

$$z \approx 2.47 \quad , \quad p \approx 0.014$$

Under L2:

$$z \approx 1.45 \quad , \quad p \approx 0.083$$

This indicates that the apparent residual dipole weakens significantly under geometry-preserving conditions.

H_0 proxy anisotropy (zHD)

Using z_{HD} , the maximum directional contrast is:

$$\Delta H_0 \approx 2.50 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

Under L1:

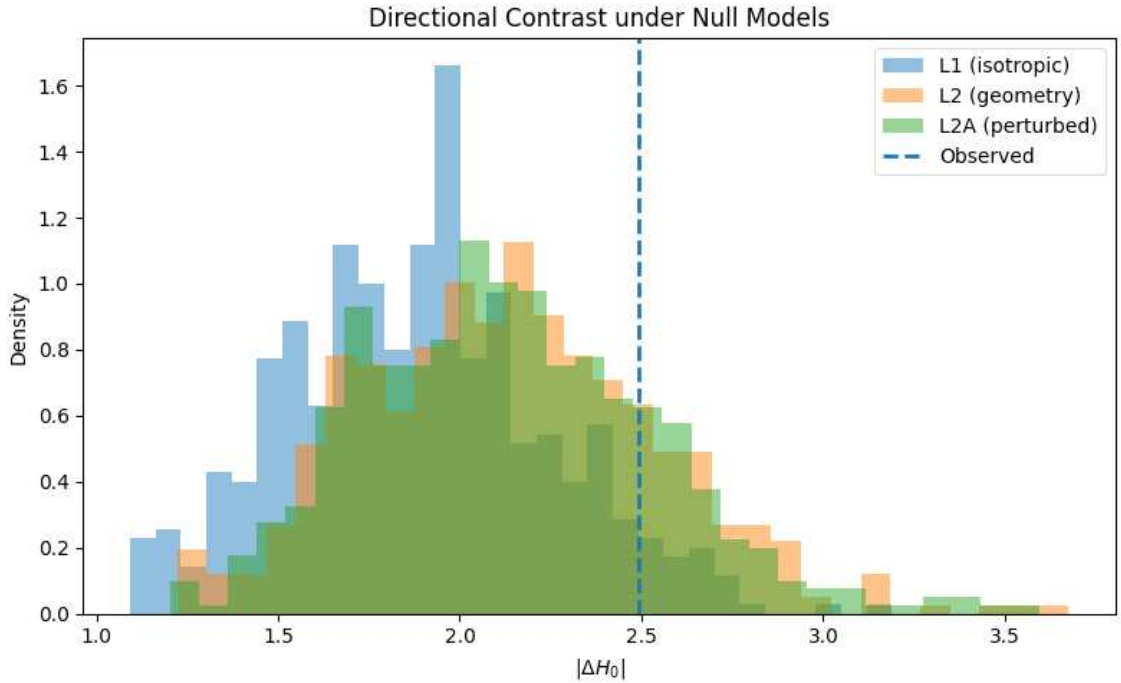
$$z \approx 1.93 \quad , \quad p \approx 0.03$$

Under L2:

$$z \approx 1.06 \quad , \quad p \approx 0.16$$

This shows that inference-level anisotropy is reduced when geometry is preserved.

The corresponding null distributions are shown in Figure 6. The observed contrast lies closer to the geometry-preserving distributions than to the naive isotropic baseline, illustrating the reduction of apparent significance under L2-type nulls.



[Figure 6]. Directional contrast of the Pantheon+ SH0ES H_0 -proxy under the AIP null hierarchy.

The observed redshift-matched maximum contrast is

$|\Delta H_0| \approx 2.50 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Under the naive isotropic null (L1), the observed value lies in the upper tail of the distribution, whereas under geometry-preserving null models (L2 and L2A) the null distributions shift toward the observed value. This indicates that the apparent direction-dependent H_0 -proxy contrast is weakened when observational geometry is preserved.

Redshift definition test (zCMB)

Using z_{CMB} , the observed contrast becomes:

$$\Delta H_0 \approx 2.19 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

with:

$$z_{L1} \approx 0.94 \quad , \quad z_{L2} \approx 0.37$$

indicating that the signal becomes statistically consistent with null expectations under all null models.

Duplicate removal (robustness test)

After removing duplicated supernova entries, the directional contrast becomes:

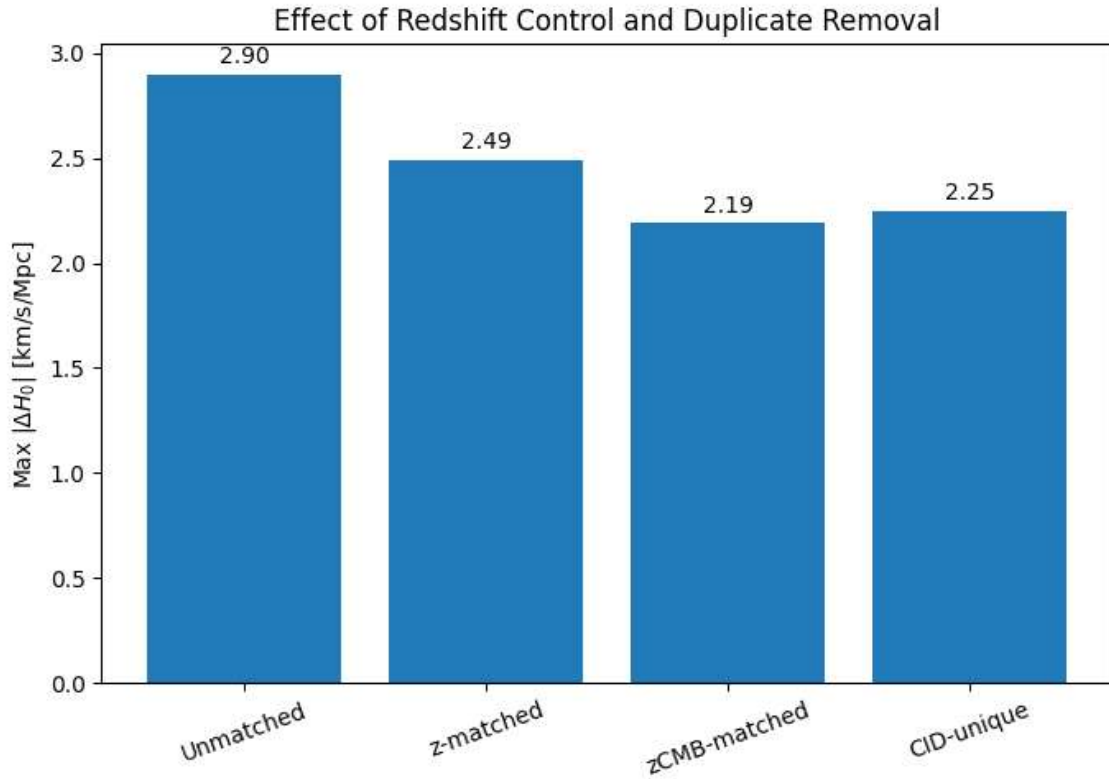
$$\Delta H_0 \approx 2.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

with:

$$p_{L1} \approx 0.33 \quad , \quad p_{L2} \approx 0.55$$

showing that the apparent anisotropy becomes fully consistent with null distributions.

The cumulative effect of these controls is summarized in Figure 7.



[Figure 7]. Effect of redshift control and duplicate removal on the maximum direction-dependent H_0 -proxy contrast.

The apparent contrast is largest in the unmatched scan ($2.90 \text{ km s}^{-1} \text{ Mpc}^{-1}$), decreases after redshift matching, weakens further under the z_{CMB} definition, and remains consistent with null expectations after duplicate supernova entries are removed. This progression indicates that the apparent signal is not a robust inference-level anisotropy, but is sensitive to observational and sample-selection effects.

►4.8.4 Interpretation

These results reveal a clear separation between residual-level directional structure and inference-level physical interpretation.

Apparent directional patterns can appear in residuals under non-uniform sky coverage and sample selection. However, once geometry-preserving and redshift-controlled null conditions are imposed, these patterns do not remain identifiable as stable direction-dependent H_0 -proxy variations.

This indicates that geometry-induced directional structure is not equivalent to physical anisotropy. In particular, apparent directional variations in inferred parameters may arise without requiring an underlying anisotropic expansion.

Therefore, residual-level anisotropy cannot be used as independent evidence for anisotropic cosmology unless identifiability is first established under geometry-preserving validation.

►4.8.5 Conclusion

The Pantheon+ analysis does not support the presence of robust direction-dependent variation in the H_0 -proxy.

Apparent signals observed under naive assumptions are reduced or eliminated under geometry-preserving and redshift-controlled conditions, and disappear entirely when duplicate entries are removed.

These results indicate that direction-dependent variations in supernova data do not provide support for anisotropic explanations of the Hubble tension, and are consistent with geometry-dependent observational effects.

These results suggest that previously reported directional variations in H_0 -related inference may require re-evaluation under geometry-preserving frameworks, as such signals may arise from observational geometry rather than intrinsic anisotropy.

In this sense, anisotropy-based interpretations of the Hubble tension may rely on signals that are not structurally identifiable within the present framework.

Therefore, apparent direction-dependent variations in H_0 -proxy estimates do not constitute independent evidence for anisotropic cosmology unless identifiability is established under geometry-preserving conditions.

This does not resolve the Hubble tension. Rather, it constrains anisotropy-based explanations of it by requiring geometry-preserving identifiability before physical interpretation.

Conclusion of Section 4

The results presented in this section form a coherent structural picture. The observed anisotropy is strongly separated from naive random baselines, but becomes statistically consistent once geometry-preserving constraints are imposed.

This consistency is robust under variations of null construction, alternative alignment statistics, bootstrap resampling, and partitioned subsets. Structure-only nulls fail to reproduce the signal, while angular-only constructions preserve it and radial-only constructions do not. Direct modification of the angular mask changes the measured anisotropy, and more structured nulls—including footprint-aware and shell-structured clustering-preserving constructions—continue to support the same interpretive pattern.

Taken together, these results show that the observed anisotropy is not a stable, geometry-independent feature of the data. Rather, within the present analysis, it emerges from the interaction between angular structure and observational geometry.

More precisely, the observed anisotropy can be generated under geometry-preserving null constructions without requiring a uniquely intrinsic structural origin.

Importantly, this behavior is not limited to structural observables. When the same framework is applied to inference-level quantities derived from supernova data, apparent directional signals are reduced or eliminated under geometry-preserving and redshift-controlled conditions. In particular, H_0 -proxy variations that appear under naive assumptions become statistically consistent with null expectations once observational geometry and sample structure are properly accounted for.

The central outcome of this section is therefore not the denial of anisotropy, but the demonstration that its physical origin is not uniquely identifiable under geometry-preserving conditions. This non-identifiability extends from structural observables to derived cosmological quantities, indicating that apparent anisotropy

cannot be treated as physically interpretable unless it remains distinguishable under geometry-preserving validation.

5. Interpretive Collapse and Non-Identifiability of Anisotropy

5.1 Interpretive Structure of Observed Anisotropy

The preceding results reveal a consistent structural pattern. The observed anisotropy signal appears as a strong deviation under the naive random null (L1), which, under simple comparison, may suggest the existence of a preferred axis.

However, once the geometry-preserving null (L2), which retains both survey footprint and radial selection, is applied, the same level of signal is reproduced.

This structure is not limited to the primary void catalog. The same pattern appears when the analysis is applied to independent datasets, and further persists in isotropic mock data where no intrinsic anisotropy exists.

Thus, the observed anisotropy is not confined to a single dataset, but appears as a reproducible structure arising from observational geometry and null construction within the present framework.

This directly implies that the observed anisotropy is not necessarily an indicator of a new physical structure, but may instead reflect a structural effect induced by observational constraints.

Accordingly, the central question shifts from “Does anisotropy exist?” to “What is the origin of the observed anisotropy?”

We summarize this structural limitation as follows:

Working description of cosmological non-identifiability

If an observed anisotropy can be reproduced with comparable statistical strength under geometry-preserving null conditions, then its physical origin cannot be uniquely identified.

In such cases, the signal is non-identifiable within the adopted observational and null-model framework, because it admits generatively equivalent explanations from both intrinsic structure and observational geometry.

This principle is not specific to a particular dataset, but reflects a structural property of anisotropy under observational constraints.

In this sense, the observed anisotropy does not constitute an independent physical observable, but a geometry-dependent statistical outcome.

5.2 Non-Identifiability of Anisotropy Detection

The most important result of this study is that the interpretation of observed anisotropy is fundamentally non-identifiable.

Under null models that do not preserve observational geometry, anisotropy induced by selection effects cannot be statistically distinguished from intrinsic physical anisotropy.

This non-identifiability is observed not only in void data, but also in independent galaxy data and isotropic mock data.

The same statistic is repeatedly reproduced under distinct generative mechanisms, preventing unique determination of the signal's origin.

In this sense, the ambiguity is fundamentally generative: identical observed patterns can arise under geometry-preserving conditions without requiring a uniquely intrinsic structural origin, limiting the physical interpretability of the signal.

The key result of this work is therefore not the presence or absence of anisotropy, but the fact that its interpretation is, in principle, non-identifiable.

This is not merely statistical uncertainty, but structural non-identifiability: distinct generative mechanisms—observational geometry and intrinsic structure—can produce indistinguishable observational outcomes within the present framework.

This conclusion is restricted to the present analysis framework and the class of geometry-preserving null constructions considered here.

Detection and interpretation are fundamentally distinct operations, and failure to distinguish them leads directly to non-identifiability.

All these observations can be summarized as a single structural statement: identical observational outcomes do not imply a unique generative origin.

This means that the problem is not simply that the signal is statistically uncertain, but that its physical source is underdetermined within the present observational framework.

We note that identifiability may be restored in regimes where angular structure is sufficiently weak or approximately isotropic, or where geometric constraints are minimized. In such cases, the distinction between intrinsic structure and observational geometry may become resolvable.

It is not a conventional selection bias problem, but a structural non-identifiability arising from generative equivalence between geometry and intrinsic signals.

This non-identifiability extends beyond structural observables to inference-level quantities. In the Pantheon+ supernova sample, apparent directional variations in H_0 -proxy estimates weaken or disappear under geometry-preserving and redshift-controlled conditions, indicating that such effects are not robust indicators of physical anisotropy.

In this sense, the problem is not that anisotropy is difficult to detect, but that its interpretation becomes fundamentally unconstrained without geometry-preserving validation.

This has direct implications for the interpretation of previously reported anisotropy signals. When null models fail to preserve observational geometry, selection-induced patterns may become indistinguishable from intrinsic structure, leading to an overestimation of statistical significance.

Within the present framework, such cases correspond to generative equivalence between geometry and signal, where multiple distinct mechanisms produce observationally identical outcomes.

5.3 Re-evaluation of Existing Anisotropy Interpretations

These results call for a systematic re-evaluation of prior anisotropy studies. In particular, anisotropy signals detected under naive null frameworks cannot exclude the possibility that they arise from selection geometry.

As a result, anisotropy claims based solely on naive null comparisons cannot be considered physically conclusive.

In this sense, the present work does not introduce a new anisotropy, but redefines the conditions under which existing interpretations remain valid.

In particular, several previously reported detections of large-scale anisotropy—such as PCA-based axis identification, hemispherical asymmetry, and pair-alignment analyses—have been evaluated primarily under naive or partially constrained null models.

Within the null hierarchy framework introduced here, such detections require systematic re-evaluation. Signals that appear statistically significant under L1 (naive random null) may become statistically indistinguishable once geometry-preserving constraints (L2) are imposed.

Therefore, previously claimed anisotropy signals should not be considered physically conclusive unless they demonstrably persist beyond geometry-preserving null conditions.

Signals that appear extraordinary under L1 may become statistically typical once geometry-preserving constraints are imposed.

This suggests that many previously reported directional signals may rely on null assumptions that do not fully capture the observational selection function.

Within the AIP framework, such analyses correspond to lower-level null constructions, in which angular structure is either simplified or removed. As a result, the statistical significance of anisotropy may be systematically overestimated when evaluated under incomplete null conditions.

Importantly, this does not invalidate prior results, but places them within a framework where their physical interpretability depends on the level of geometric consistency in the analysis.

The critical distinction is not whether anisotropy is detected, but whether it remains interpretable under geometry-preserving constraints.

From this perspective, existing anisotropy results must be reinterpreted according to the structure of the null model employed. Analyses based on naive random nulls correspond to the L1 level in this framework, under which selection-induced anisotropy cannot be distinguished from intrinsic structure.

Only when geometry-preserving nulls are applied can a signal pass the threshold of physical interpretability.

Accordingly, anisotropy claims that do not demonstrate persistence under geometry-preserving nulls should be regarded as non-identifiable rather than physically established.

5.4 Minimal Conditions for Physical Interpretability

This analysis does not deny the existence of anisotropy. Instead, it establishes the minimal conditions required for physical interpretation.

First, the signal must remain distinguishable from geometry-preserving null distributions.

Second, the signal must remain stable under reasonable variations of null construction.

Third, the signal must persist under data partitioning, including size, distance, and other relevant observational subdivisions.

Fourth, the signal must not be fully reproducible by geometry-only or structure-only proxy constructions that lack the proposed intrinsic physical mechanism.

Geometry-preserving validation is therefore a necessary, though not sufficient, condition for assigning a physical origin to an anisotropy signal.

5.5 Fundamental Questions of Interpretation

These results raise several fundamental questions:

1. Can a signal that does not persist beyond geometry-preserving validation be interpreted as a physical structure?
2. What does anisotropy detected only under naive null conditions actually measure?
3. If selection-induced and intrinsic anisotropy cannot be distinguished, can the current methodology support physical interpretation?
4. Can analyses that do not control for geometry justify physical conclusions?
5. If the same signal can be reproduced by geometry-preserving nulls, is an additional physical explanation required?
6. Under what conditions can anisotropy be considered a uniquely physical observable?

5.6 Concluding Statement

The observed anisotropy is not direct evidence of an independent physical structure, but a signal that can arise from the coupling of observational geometry and selection effects.

Thus, the signal cannot be uniquely attributed to intrinsic cosmic structure and remains statistically compatible with a geometry-induced origin. Within the present framework, no statistical evidence requires a uniquely physical origin.

This work does not propose a new cosmic structure. Instead, it defines the conditions under which a signal can carry physical meaning.

We redefine not the universe, but the conditions under which its structure can be meaningfully interpreted.

This is not the end of anisotropy studies, but the beginning of a framework in which their interpretability can be rigorously defined.

6. AIP Framework:

Testable Conditions for Interpreting Cosmic Anisotropy

This study does not aim to explain a specific anisotropy signal, but to establish the conditions under which such a signal can carry physical meaning.

In this sense, the AIP is not merely a heuristic guideline, but a structural validation condition for assigning physical meaning to anisotropy within the present framework.

The criteria presented in this section define the minimal requirements for evaluating the interpretability of observed anisotropy. They are not limited to a particular dataset, but are formulated as a general analytical framework.

The following conditions do not constitute a result, but a methodological standard for determining interpretability.

In what follows, these conditions are collectively referred to as the **Anisotropy Identifiability Protocol (AIP)**.

In the present work, the AIP is not introduced as an abstract formalism alone, but is grounded in the empirical results obtained above, including robustness under null variations, alternative alignment statistics, bootstrap resampling, partition stability, angular-radial decomposition, mask sensitivity, and structured mock constructions.

6.1 Geometry Removal Condition

If angular footprint and radial selection are removed or sufficiently symmetrized, the anisotropy should be strongly reduced relative to the observed level.

This condition provides the most basic criterion for distinguishing selection-induced anisotropy. Signals generated by observational geometry should disappear when geometry is removed or symmetrized.

Conversely, anisotropy with a candidate physical origin should remain distinguishable, at least partially, under such transformations.

6.2 Geometry-Only Reproducibility Condition

This condition tests whether anisotropy of comparable strength can be reproduced using only selection geometry, without invoking intrinsic structure. As demonstrated in this study, if the same anisotropy pattern is reproduced in isotropic mock data, the signal cannot be uniquely attributed to a physical origin.

In the present analysis, this condition is supported not only by the original geometry-preserving null, but also by footprint-aware and shell-structured clustering-preserving constructions, which recover a substantial fraction of the signal or remain statistically consistent with the observed anisotropy under more structured assumptions.

In other words, a signal reproducible by geometry alone cannot be treated as requiring a uniquely intrinsic physical explanation.

Such signals therefore fail to satisfy the minimal criterion for physical interpretability.

6.3 Null-Dependence Condition

This condition evaluates whether the significance of anisotropy depends strongly on the choice of null model.

If the statistical significance of anisotropy varies substantially with the null model, the signal becomes non-identifiable within the adopted null hierarchy.

In particular, signals that appear only under naive nulls but weaken or vanish under geometry-preserving nulls cannot be uniquely attributed to intrinsic structure.

This condition is strengthened when the same interpretive pattern remains stable under reasonable variations of null construction, including perturbed angular realizations and structured proxies, while collapsing under isotropic angular degradation.

This behavior is also observed in inference-level quantities derived from supernova data, where apparent anisotropy is reduced or eliminated when evaluated under geometry-preserving null conditions. This suggests that the AIP framework applies not only to structural observables but also to cosmological inference.

6.4 Interpretability Criterion

Anisotropy without geometry-preserving validation cannot be treated as uniquely physically interpretable within the present framework.

Moreover, interpretability should not depend on a single alignment statistic or a single realization of the catalog. In the present study, the same conclusion remains stable across alternative alignment measures, bootstrap resampling, and partitioned subsets, indicating that interpretability must be assessed structurally rather than by a single nominal detection.

This condition summarizes the central conclusion of the present work.

Anisotropy detected without preserving observational geometry cannot be treated as physically interpretable and must pass geometry-preserving validation prior to interpretation.

6.5 Necessary vs Sufficient Condition

Geometry-preserving validation is a necessary, though not sufficient, condition for establishing a physical origin.

Analyses that do not control for geometry cannot, by themselves, serve as a sufficient basis for physical interpretation. However, satisfying geometry-preserving conditions does not automatically guarantee a physical origin.

This condition defines the minimal requirement for interpretability, while additional validation remains necessary.

Such additional validation may include robustness under alternative null constructions, stability across alignment statistics and sample realizations, decomposition of angular and radial contributions, and direct sensitivity tests under modified observational masks. In this sense, geometry-preserving validation is only the starting point of interpretation, not its endpoint.

Summary of Section 6

The AIP framework defines a minimal validation standard for interpreting cosmic anisotropy. Its purpose is not to explain a specific signal, but to determine whether a detected directional pattern can be assigned a physically meaningful origin.

Within this framework, a signal must remain distinguishable under geometry-preserving null conditions, remain stable under reasonable variations of null construction, and resist reproduction by geometry-only or structure-only proxy constructions. These requirements are supported in the present study by a convergent set of empirical tests: null-variant robustness, metric-independence, bootstrap stability, partition consistency, angular-radial decomposition, mask sensitivity, and structured mock agreement.

Geometry-preserving validation is therefore a necessary, though not sufficient, condition for physical interpretation. Signals that fail this validation may remain statistically detectable, but they cannot be treated as physically identifiable within the present framework.

Accordingly, the AIP does not eliminate anisotropy studies. It defines the conditions under which anisotropy claims become interpretable. Future analyses of directional effects in cosmological data should therefore incorporate geometry-preserving null models as a baseline requirement.

It should be emphasized that consistency with a geometry-preserving null does not prove the absence of intrinsic anisotropy. If an intrinsic anisotropic signal is perfectly degenerate with the observational geometry, L2 may absorb that signal into the reference distribution. In such a case, however, the appropriate conclusion is not physical absence, but non-identifiability within the present statistic and null hierarchy.

Full cosmological N-body mocks would be required to test whether a specific physical model can generate an anisotropy that remains distinguishable beyond geometry-preserving null conditions. The present work does not claim to exclude such model-dependent possibilities. Its conclusion is more limited: without geometry-preserving validation, statistical anisotropy alone cannot be assigned a uniquely physical origin.

7. Conclusion

This study shows that the observed cosmic anisotropy pattern is not direct evidence of a new physical structure, but can be reproduced as a geometric effect arising from the combination of survey footprint and radial selection.

Under naive random null conditions, strong anisotropy appears. However, once geometry-preserving nulls are applied, the same level of signal is reproduced, and this structure remains stable even after data partitioning.

These results are consistently reproduced not only in void data, but also in independent galaxy datasets and isotropic mock data.

The same structural pattern is also reproduced in an independent void catalog under an identical analysis pipeline, further supporting the conclusion that the result is not tied to a specific dataset or catalog construction.

This indicates that the observed anisotropy is not confined to a single dataset or catalog construction, but is reproducible under observational geometries considered within the present framework.

In other words, the same statistic can be simultaneously explained by distinct generative mechanisms—intrinsic structure and observational geometry—and therefore the physical origin of the signal cannot be uniquely determined.

Within the present framework, no statistical evidence requires invoking a uniquely intrinsic cosmic origin for the observed anisotropy.

This implication extends beyond structural observables to inference-level quantities. When the same framework is applied to supernova-derived H_0 proxies in the Pantheon+ dataset, apparent direction-dependent variations weaken or disappear under geometry-preserving and redshift-controlled conditions.

In particular, marginal anisotropy signals observed under naive assumptions are not sustained once observational geometry and sample structure are properly accounted for, and become statistically consistent with null expectations after duplicate removal.

This indicates that inference-level anisotropy in the H_0 -proxy is not robust within the present analysis, and remains consistent with geometry-dependent observational effects.

This work does not deny the existence of anisotropy. Instead, it clarifies the conditions under which anisotropy can be physically interpreted. Detection and interpretation are fundamentally distinct operations, and the former does not guarantee the latter.

Accordingly, detection alone cannot be taken as evidence of intrinsic structure without satisfying identifiability conditions.

Geometry-preserving validation is therefore a necessary, though not sufficient, condition for establishing a physical origin.

This conclusion is supported by consistent results across null-variant robustness tests, alternative alignment statistics, bootstrap resampling, partition stability, angular-radial decomposition, and mask sensitivity analyses.

This study also provides a testable framework. In datasets with similar observational geometry, naive null analyses are expected to reproduce comparable anisotropy signals, while geometry-preserving nulls will either eliminate such signals or render them statistically indistinguishable.

The criteria presented in Section 6 formalize the conditions under which anisotropy can be considered physically meaningful.

In this sense, the contribution of this work is not the proposal of a new cosmic axis, but the establishment of criteria for determining when a signal resembling one can be meaningfully inferred.

Detection does not guarantee interpretation.

If anisotropy signals of comparable strength can emerge purely from observational geometry, then a critical question must be confronted:

To what extent have we been interpreting the structure of the universe, and to what extent have we been interpreting the structure of our observations?

This work suggests that, without geometry-preserving validation, the two may be fundamentally indistinguishable.

We did not identify a new cosmic axis: we showed how a signal resembling one can emerge from observational geometry.

This also raises a constructive question: what conditions would be required to restore identifiability?

In principle, breaking the degeneracy would require either:

- (i) independent constraints on the selection function,
- (ii) datasets with near-uniform angular coverage, or
- (iii) observables that are invariant under geometry-preserving transformations.

Future observational strategies should therefore prioritize geometric completeness and independent calibration of selection effects.

From a methodological perspective, overcoming projection-induced limitations requires explicitly incorporating observational geometry into the inference process. In this context, anisotropy should be evaluated relative to geometry-preserving reference distributions, consistent with the ALP framework proposed in this work.

More generally, combining observations across multiple effective perspectives, including independent datasets and lensing-induced projections, may help reduce degeneracies introduced by fixed observational geometry.

Finally, it may be advantageous to focus on quantities that are less sensitive to projection effects, such as topology-based or invariant measures, which could provide complementary constraints in regimes where conventional alignment statistics become non-identifiable.

In particular, this work shows that direction-dependent explanations of cosmological discrepancies, including those proposed in the context of the Hubble tension, must satisfy identifiability conditions before being considered physically meaningful.

Without such validation, statistical significance alone does not confer physical interpretability, and anisotropic signatures cannot be regarded as independent evidence for cosmological models.

This result places a direct constraint on a class of proposed solutions to cosmological tensions: explanations based on directional effects must first demonstrate identifiability under geometry-preserving conditions.

We redefine not the universe, but the conditions under which it is interpreted.

Appendix A. Methodological Details

Notation follows the convention used in the main text. For typographical consistency, ordinary italic symbols are used throughout instead of boldface or blackboard-style notation. Vector and scalar meanings are defined contextually: $\mathbf{X}$ denotes position, $\mathbf{R}_{ij}$ denotes the pair-separation vector, $\mathbf{n}$, $\mathbf{u}$, and $\hat{\mathbf{x}}$ denote unit directions, and $R = |\mathbf{X}|$ denotes scalar radial distance.

A.1 Anisotropy Statistic

The anisotropy statistic used throughout this work is defined as

$$A = \langle |\cos \theta_{ij}| \rangle$$

where θ_{ij} is the angle between the pair-separation vector $\mathbf{R}_{ij} = \mathbf{X}_j - \mathbf{X}_i$ and the fixed global axis $\mathbf{n}$.

For each pair of points (i, j) , the unit separation vector is defined as

$$\hat{R}_{ij} = \frac{X_j - X_i}{|X_j - X_i|}$$

The directional alignment with the fixed global axis n is then measured by

$$|\cos \theta_{ij}| = |\hat{R}_{ij} \cdot n|$$

The statistic A is obtained by averaging this quantity over all sampled pairs.

This definition directly captures directional alignment while avoiding additional model-dependent assumptions, ensuring minimality and robustness.

For an isotropic distribution, the expected value of A is **0.5**. This follows because $\cos \theta$ is uniformly distributed over $[-1, 1]$, and therefore $|\cos \theta|$ is uniformly distributed over $[0, 1]$:

$$E[|\cos \theta|] = \int_0^1 x dx = \frac{1}{2}$$

where $E[\bullet]$ denotes expectation.

This follows from the isotropy of the distribution, which implies a uniform distribution of directions on the sphere.

Thus, deviations from 0.5 may arise either from intrinsic anisotropy or from observational constraints imposed by the survey geometry.

A.2 Null Model Constructions

Three classes of null models are used to evaluate the origin and interpretability of the observed anisotropy.

►L1: Naive Random Null

In the L1 construction, point positions are uniformly randomized within the bounding volume of the dataset.

This model preserves neither radial distributions nor survey geometry, and serves as a baseline reference for unconstrained randomness.

►L2: Geometry-Preserving Null

In the L2 construction, the radial distances of the original points are preserved, while angular directions are resampled from the empirical distribution of unit vectors.

Specifically, for each point:

- The radial distance $R = |X|$ denotes the Euclidean norm of X .
- A new direction u is sampled from the set of observed directions.
- The new position is constructed as $X' = Ru$.

This procedure preserves the survey footprint and selection geometry while removing intrinsic structural correlations.

It should be noted that this construction is not equivalent to simple angular randomization (e.g., ϕ -shuffling).

The use of empirical direction resampling preserves the observational footprint and introduces correlations in the resulting distribution that are absent in simplified implementations.

As a result, while the mean anisotropy level remains consistent, the dispersion of the statistic may vary depending on the exact implementation details, even when the underlying construction principle is preserved.

This construction corresponds to a geometry-preserving null that maintains the observational measure of the dataset, including radial selection and empirical angular footprint, as formalized in Appendix B.

This distinction is structural. L1 samples from an unconstrained spatial domain, whereas L2 samples from a constrained observational manifold defined by the survey footprint and radial selection.

As a result, L1 and L2 do not represent two numerical variants of the same null hypothesis, but different generative spaces. Therefore, statistical significance measured relative to L1 does not imply physical significance when the observational geometry is non-uniform.

This distinction is not merely technical, but determines the interpretability of the resulting statistic.

►L3: Structure-Only Null

In the L3 construction, the radial or structural distribution of the dataset is preserved while the empirical angular footprint is removed or randomized.

This isolates the contribution of intrinsic structure independent of observational geometry, allowing assessment of whether structure alone is sufficient to reproduce the observed anisotropy.

In addition to these baseline constructions, extended null variants are also considered in the main analysis, including perturbed angular realizations (L2A), isotropic angular randomization (L2B), footprint-constrained sampling, and

shell-structured clustering-preserving proxies. These extensions are not separate model classes, but operational variations used to test robustness within the geometry-preserving framework.

A.3 Pair Sampling Procedure

The anisotropy statistic is computed using randomly sampled point pairs.

For each realization:

- A large number of pairs, typically about 10^5 , is drawn.
- Pairs are selected with replacement using uniform random indices.
- Pairs with zero separation are excluded.

This procedure ensures statistical stability while avoiding systematic biases.

The robustness of the resulting statistic is further verified through bootstrap resampling and partitioned subset analyses, confirming that the observed consistency is not driven by specific pair selections or localized data subsets.

Symmetry is preserved and double counting is implicitly handled through random sampling.

Since the statistic $\mathcal{A}$ depends only on directional alignment and not on absolute position, the sampling procedure does not introduce directional bias provided that pair selection is uniform.

The use of a large random pair sample ensures convergence of the estimator toward its expected value, while avoiding dependence on any particular local subset of the catalog.

A.4 Global Axis Definition

The global axis $\mathcal{n}$ is fixed throughout the analysis and is not re-estimated under null realizations.

This ensures that the anisotropy statistic does not artificially encode alignment through axis construction, and that all null comparisons are performed under a consistent directional reference.

Fixing the global axis ensures that alignment is evaluated relative to a consistent reference direction across all null constructions.

However, this also means that any preferred direction already encoded in the dataset or in the observational geometry can be reflected in the statistic. The axis does not by itself introduce anisotropy; rather, it exposes directional structure already present in the sampling domain.

A.5 Implementation Notes

All computations are deterministic given the input dataset and random seed

initialization.

No parameter tuning, bin optimization, or model-dependent adjustments are applied in the definition of the statistic or the construction of the null models.

This absence of parameter tuning ensures that the observed anisotropy is not a result of model optimization, but follows directly from the dataset and the defined geometric constraints.

Consequently, the interpretation is not driven by overfitting or parameter-induced bias.

The analysis is therefore fully reproducible in principle from the definitions and procedures described above.

Additional validation steps, including alternative alignment statistics, angular-radial decomposition, and mask sensitivity tests, are performed in the main text and follow directly from the same deterministic framework defined here. These procedures do not introduce additional model assumptions, but extend the same computational structure to assess interpretability and robustness.

The same framework extends directly to inference-level quantities without introducing additional model assumptions.

A.6 Inference-Level Extension: H_0 Proxy Construction

To extend the non-identifiability framework to inference-level quantities, we define a directional H_0 -proxy derived from supernova observables.

The luminosity distance d_L is computed from the distance modulus:

$$\mu = 5 \log_{10}(d_L) + 25$$

which yields:

$$d_L = 10^{(\mu - 25)/5}$$

Using this, we define the H_0 -proxy as:

$$H_0^{\text{proxy}} = \frac{cz}{d_L}$$

where z denotes the redshift, taken as either z_{HD} or z_{CMB} .

This definition corresponds to a local Hubble-law proxy, where the ratio cz/d_L serves as an effective estimator of the expansion scale for each object.

The use of both redshift definitions provides a consistency check against systematic effects associated with peculiar velocity corrections and reference frame choices.

Because the directional partition is defined using sky position and a scanned axis, any geometry-induced alignment in the angular distribution can propagate into the inferred directional contrast. The purpose of the H_0^{proxy} construction is therefore not to measure an absolute Hubble constant, but to test whether apparent direction-dependent inference survives geometry-preserving validation.

► Directional contrast

Given a unit axis $\hat{n}$, the angular alignment for each data point is:

$$\cos \theta_i = \hat{x}_i \cdot \hat{n}$$

where $\hat{x}_i$ is the unit direction vector of the i -th object.

The dataset is partitioned into two subsets based on the magnitude of $|\cos \theta|$:

- parallel subset: largest values
- perpendicular subset: smallest values

The directional contrast is defined as:

$$\Delta H_0 = \langle H_0^{\text{proxy}} \rangle_{\parallel} - \langle H_0^{\text{proxy}} \rangle_{\perp}$$

► Redshift-controlled estimator

To eliminate redshift-dependent bias,
the dataset is partitioned into bins in redshift:

$$z \in [z_k, z_{k+1}]$$

Within each bin, the directional partition is performed independently,
and the resulting contrasts are averaged:

$$\Delta H_0^{(z\text{-matched})} = \frac{1}{N_{\text{bins}}} \sum_k \Delta H_0^{(k)}$$

► Duplicate removal

To avoid artificial weighting due to repeated entries,
objects sharing identical identifiers are merged:

$$\overline{H}_0 = \frac{1}{N_i} \sum_{j \in i} H_{0,j}$$

where i indexes unique objects and j indexes duplicate entries.

► Null application

The same null hierarchy (L1, L2, L2A) is applied to H_0^{proxy} by:

- randomizing angular positions (L1)

- preserving geometry while permuting values (L2)
- perturbing geometry (L2A)

This ensures consistency with the structural anisotropy analysis.

► Interpretation

This construction allows direct testing of whether directional patterns observed in supernova data propagate into inference-level quantities.

If ΔH_0 becomes statistically consistent with null distributions under geometry-preserving conditions, the apparent anisotropy is interpreted as non-identifiable at the inference level.

This ensures that the extension to inference-level quantities remains consistent with the structural non-identifiability framework established for anisotropy statistics.

Therefore, consistency with geometry-preserving null models at the level of ΔH_0 indicates that the apparent directional variation cannot be independently attributed to intrinsic physical anisotropy within the present framework.

Appendix B: Computational Procedure and Reproducibility

B.1 Overview

The analysis consists of three main stages:

1. Construction of the dataset and determination of the global axis
2. Computation of anisotropy statistics
3. Generation of null distributions and statistical comparison

The framework is designed not merely to detect anisotropy, but to evaluate how the same statistic behaves under different generative conditions.

This procedural decomposition ensures that each component of the analysis can be independently verified and reproduced.

B.2 Dataset and Global Axis

The dataset consists of a catalog of objects with spatial positions and associated structural properties.

A global axis is derived directly from the dataset using principal component

analysis (PCA). The axis is determined from the data itself and is not externally imposed.

To ensure that the axis does not artificially induce alignment, its stability is verified through independent recomputation and consistency checks. The angular difference between the original axis and the recomputed axis is negligible, supporting internal consistency.

This supports the interpretation that the axis is a stable feature of the dataset rather than a numerical artifact of a particular realization.

B.3 Anisotropy Statistics

For each pair of objects, the separation vector is computed. The alignment between this vector and the global axis is quantified.

Formally, the primary statistic is defined as:

$$A = \langle |\cos \theta_{ij}| \rangle$$

where θ_{ij} is the angle between the pair-separation vector of objects i and j and the global axis, and the average is taken over all sampled pairs.

This statistic captures directional alignment while avoiding additional model assumptions or free parameters.

To test robustness against the choice of metric, alternative statistics are also considered, including:

- $\langle \cos^2 \theta \rangle$

- A mean-centered deviation measure $\langle |\cos \theta - \langle \cos \theta \rangle| \rangle$

These statistics provide complementary measures of alignment while preserving the same geometric interpretation.

B.4 Pair Sampling Procedure

The anisotropy statistics are, in principle, defined over all unique object pairs.

In practice, a large random subset of pairs is used to ensure computational efficiency while maintaining statistical stability. The number of sampled pairs is typically on the order of 10^5 .

Sampling is performed uniformly over the set of all possible pairs, ensuring unbiased estimation.

This approach preserves statistical accuracy while avoiding redundant computations.

B.5 Null Model Construction

Three classes of null models are considered:

- **L1 (Naive Random Null):**

Positions are randomized without preserving observational geometry.

- **L2 (Geometry-Preserving Null):**

Transformations preserve both the angular coverage (survey footprint) and the radial distribution.

In this sense, L2 can be interpreted as a measure-preserving transformation with respect to the observational distribution.

In practice, this is implemented by preserving the radial distances of objects while resampling angular directions within the observed footprint.

In inference-level applications, this is extended by preserving the observed spatial positions while randomly permuting associated scalar values, such as the H_0 -proxy, across fixed positions.

Formally, L2 can be interpreted as a measure-preserving transformation on the observational domain, ensuring that the empirical angular and radial distributions remain invariant under resampling.

- **L3 (Structure-Only Null):**

Angular positions are randomized while preserving radial structure, isolating the contribution of intrinsic spatial distribution.

These null models allow systematic comparison of the same statistic under distinct generative mechanisms.

◆B.5.1 Algorithmic Implementation of the Geometry-Preserving Null

To ensure reproducibility, the geometry-preserving null (L2) is implemented through the following procedure:

- ▶ **Step 1: Fix the survey footprint**

The angular coverage of the dataset is preserved, ensuring that all generated points lie within the observed survey region.

- ▶ **Step 2: Preserve the radial selection function**

The radial distances of objects are retained, maintaining the original radial distribution of the dataset.

- ▶ **Step 3: Apply constrained angular resampling**

Angular positions are resampled under the constraint of the survey footprint, either by empirical direction resampling or rejection sampling within the footprint.

► Step 4: Recompute the anisotropy statistic

For each realization, the anisotropy statistic is recomputed using the same pair sampling procedure and axis definition as in the observed data.

The null construction preserves angular coverage, radial selection, and sampling density under the survey footprint. As a result, the generated realizations are evaluated under the same observational constraints as the original dataset, reducing discrepancies due to domain mismatch.

B.6 Extended Validation Procedures

To ensure robustness of the results, additional validation procedures are performed:

- Perturbed angular realizations
- Fully isotropic angular randomization
- Footprint-constrained sampling
- Shell-structured clustering-preserving mock construction
- Bootstrap resampling
- Partitioned subset analysis
- Angular-radial decomposition
- Mask sensitivity tests

These procedures do not introduce new model assumptions, but instead test the stability of the results under controlled variations of the null construction and data configuration.

The consistency of results across these variations indicates that the observed behavior is not driven by a single modeling choice.

B.7 Reproducibility and Determinism

All computations are deterministic given the same input dataset and random seed.

All stochastic procedures are controlled by fixed random seeds, ensuring that identical inputs yield identical outputs across independent runs.

The full analysis pipeline can be reproduced by following the procedural steps described above.

The methodology is designed to be directly applicable to other observational datasets with similar geometric constraints, enabling independent verification.

Implementation details are available upon reasonable request.

Minor implementation variations do not significantly affect the qualitative structure of the result, as verified by variant constructions (e.g., L2A).

Together, these checks indicate that the qualitative conclusions are not driven by stochastic fluctuations or a single implementation choice. The same computational structure is then extended to inference-level quantities in Section B.8 without introducing additional model assumptions.

B.8 Extension to Inference-Level Analysis (H_0 Proxy)

To extend the AIP framework to inference-level quantities, the following computational procedure is applied to supernova datasets.

This procedure is designed to test whether direction-dependent signals observed at the residual level propagate into inference-level quantities under the same geometric constraints.

► Step 1: Construction of the H_0 proxy

For each object:

$$d_L = 10^{(\mu - 25)/5}$$

$$H_0^{\text{proxy}} = \frac{cz}{d_L}$$

where z is taken as either z_{HD} or z_{CMB} .

► Step 2: Directional scanning

A set of trial axes $\hat{n}$ is generated by uniform sampling over the unit sphere.

For each axis:

$$\cos \theta_i = \hat{x}_i \cdot \hat{n}$$

Objects are partitioned into:

- parallel subset (largest $|\cos \theta|$)
- perpendicular subset (smallest $|\cos \theta|$)

► Step 3: Directional contrast

$$\Delta H_0^{\text{proxy}} = \langle H_0^{\text{proxy}} \rangle_{\parallel} - \langle H_0^{\text{proxy}} \rangle_{\perp}$$

The maximum absolute contrast is recorded across all scanned axes.

►Step 4: Redshift-controlled estimator

To remove redshift bias:

- data are partitioned into bins in z
- directional contrast is computed within each bin
- results are averaged across bins

$$\Delta H_0^{\text{proxy}}(z\text{-matched}) = \frac{1}{N_{\text{bins}}} \sum_k \Delta H_0^{\text{proxy}}(k)$$

►Step 5: Duplicate removal

To avoid artificial weighting due to repeated entries, objects with identical identifiers are merged by averaging their H_0^{proxy} values over repeated entries.

►Step 6: Null model application

The null hierarchy is applied as follows:

- L1: randomize angular positions
- L2: preserve spatial positions while randomly permuting H_0^{proxy} values across fixed coordinates, ensuring that geometry is retained while breaking any intrinsic association between values and directions
- L2A: apply small perturbations to positions

Each realization repeats Steps 2-4.

►Step 7: Statistical evaluation

For each null ensemble:

- mean and variance of $\Delta H_0^{\text{proxy}}$ are computed
- z-scores and tail probabilities are evaluated

►Interpretation

If the observed $\Delta H_0^{\text{proxy}}$ is consistent with geometry-preserving nulls, the directional signal is considered non-identifiable at the inference level.

This ensures consistency with the structural non-identifiability framework established for anisotropy statistics.

This indicates that geometry-induced structure at the spatial level may influence inference-level quantities through the same directional partitioning.

Appendix C: Geometric Origin of Apparent Anisotropy

Notation note. In this appendix, symbols such as x , n , u , r , A , W , P , and E denote mathematical variables or operators. Subscripts such as *obs*, *true*, *iso*, *ang*, *rad*, *L1*, *L2*, and *proxy* are used as descriptive labels rather than independent variables. For typographical simplicity, these subscripts may appear in italic form depending on the equation editor, but their meaning is label-based throughout the text. Likewise, x denotes a three-dimensional position vector unless otherwise stated, n denotes a unit direction axis, and $E[\cdot]$ denotes the expected value operator. The expectation operator is written as $E[\cdot]$ instead of the blackboard-bold symbol throughout this appendix.

C.1 Conceptual Basis

Apparent anisotropy can arise from the geometry of the observational sampling itself, even in the absence of intrinsic physical anisotropy.

Let the underlying distribution of objects be statistically isotropic in three-dimensional space. The observed catalog, however, is not drawn from the full isotropic distribution, but from a restricted observational domain determined by the survey footprint, angular mask, radial selection, and sampling completeness.

We represent this observational constraint by a selection function

$$W(x) = W(r, \hat{x}) \quad (\text{C.1})$$

where $r = |x|$ is the radial coordinate and $\hat{x}$ is the angular direction on the sky. The observed distribution can then be written schematically as

$$P_{\text{obs}}(x) = \frac{W(x) P_{\text{true}}(x)}{\int W(x) P_{\text{true}}(x) d^3x} \quad (\text{C.2})$$

If the underlying distribution is isotropic, then $P_{\text{true}}(x) = P_{\text{iso}}(r)$.

In that case,

$$P_{\text{obs}}(x) = \frac{W(r, \hat{x}) P_{\text{iso}}(r)}{\int W(r, \hat{x}) P_{\text{iso}}(r) d^3x} \quad (\text{C.3})$$

Equation (C.3) shows that even when the true distribution is isotropic, the observed distribution inherits the angular and radial structure of the selection function W . Therefore, anisotropy in the observed catalog can be generated by the observational domain itself.

The alignment statistic used in this work is

$$A(\hat{n}) = \langle |\cos\theta| \rangle \quad (\text{C.4})$$

where θ is the angle between the pair-separation direction and a fixed axis $\hat{n}$. More explicitly, for two observed points x_i and x_j

$$\hat{u}_{ij} = \frac{x_i - x_j}{|x_i - x_j|} \quad (\text{C.5})$$

and

$$A(\hat{n}) = \langle |\hat{u}_{ij} \cdot \hat{n}| \rangle \quad (\text{C.6})$$

For a fully isotropic distribution over the full sphere, the expectation value is

$$E[A] = \int_0^1 \mu d\mu = \frac{1}{2} \quad (\text{C.7})$$

where $\mu = |\cos \theta|$.

However, under a non-uniform selection function, the expectation value becomes

$$E_W[A(\hat{n})] = \int |\hat{u}_{12} \cdot \hat{n}| P_W(x_1) P_W(x_2) d^3x_1 d^3x_2 \quad (\text{C.8})$$

where P_W denotes the observed distribution after selection by W .

Here, x_1 and x_2 denote three-dimensional position vectors, and $d^3x_1 d^3x_2$ denotes integration over the corresponding three-dimensional observational domain.

Equation (C.8) is central. It shows that the expected value of the anisotropy statistic is no longer determined solely by the intrinsic distribution. It is also determined by the geometry of the observational domain.

This dependence is purely geometric: it arises from the integration domain rather than from any intrinsic physical property of the underlying distribution. In particular, when the intrinsic distribution is isotropic, the directional dependence of $A(\hat{n})$ can arise from the angular and radial structure of $W(x)$.

Therefore, a deviation of A from $1/2$ is not, by itself, evidence of intrinsic anisotropy. It may instead reflect the fact that the statistic is being evaluated over a geometrically restricted observational domain.

C.2 Toy Model Demonstration

Consider an intrinsically isotropic distribution of points in three-dimensional space. If all directions are equally observable, then the pair-separation directions are isotropically distributed, and the expected alignment statistic satisfies

$$E[A] = \frac{1}{2} \quad (\text{C.9})$$

Now impose a non-uniform observational mask. For example, suppose that objects are observed only within a restricted angular region Ω_W .

The selection function may be written as

$$W(\hat{x}) = \begin{cases} 1, & \hat{x} \in \Omega_W \\ 0, & \hat{x} \notin \Omega_W \end{cases} \quad (\text{C.10})$$

The observed angular distribution is then no longer isotropic, even though the underlying distribution remains isotropic.

The expectation value of the alignment statistic becomes

$$E_W[A(\hat{n})] = \frac{\int_{\Omega_W} \int_{\Omega_W} |\hat{u}_{12} \cdot \hat{n}| d\Omega_1 d\Omega_2}{\int_{\Omega_W} \int_{\Omega_W} d\Omega_1 d\Omega_2} \quad (\text{C.11})$$

Unless Ω_W is rotationally symmetric with respect to all directions, the value of $E_W[A(\hat{n})]$ depends on the axis $\hat{n}$. In other words,

$$E_W[A(\hat{n})] \neq \frac{1}{2} \quad (\text{C.12})$$

in general.

This shows that directional alignment can emerge solely from the restriction of the integration domain. No intrinsic physical anisotropy is required.

The same logic applies to a finite survey volume. If points are uniformly distributed inside an elongated or asymmetric observational volume V , then the pair-separation vectors are not isotropically distributed. They are biased toward the longer dimensions of the survey domain. Hence, the statistic $A(\hat{n})$ is enhanced along axes associated with the dominant geometry of V .

This can be represented schematically by the shape tensor of the observational domain,

$$S_{ij} = \int_V x_i x_j d^3x \quad (\text{C.13})$$

The eigenvectors of S_{ij} define the principal directions of the observational volume. If the survey geometry is anisotropic, then these directions naturally act as preferred directions for pair-alignment statistics.

This does not imply that every detected axis is identical to a shape-tensor eigenvector. Rather, it shows that an anisotropic observational domain possesses its own geometric principal directions, and alignment statistics can respond to those directions even when the underlying physical distribution is isotropic.

Thus, in this setting, anisotropy can arise from the imposed selection geometry rather than from the underlying physical distribution alone. The emergence of directional alignment can therefore be understood as a consequence of the

sampling constraint within the specified observational domain.

C.3 Relation to Observational Footprint

In real cosmological datasets, the selection function is not uniform. The observed catalog is shaped by survey strategy, sky coverage, masking, depth variation, redshift selection, and completeness limits.

This can be decomposed schematically as

$$W(r, \hat{x}) = W_{\text{rad}}(r) W_{\text{ang}}(\hat{x}) \quad (\text{C.14})$$

where $W_{\text{rad}}(r)$ describes radial selection and $W_{\text{ang}}(\hat{x})$ describes angular footprint.

For an intrinsically isotropic distribution,

$$P_{\text{true}}(x) = P_{\text{iso}}(r) \quad (\text{C.15})$$

the observed distribution becomes

$$P_{\text{obs}}(r, \hat{x}) \propto W_{\text{rad}}(r) W_{\text{ang}}(\hat{x}) P_{\text{iso}}(r) \quad (\text{C.16})$$

Equation (C.16) shows that the observed catalog necessarily inherits the angular structure of W_{ang} . Therefore, the sky footprint is not a secondary observational detail. It is part of the generative structure of the measured statistic.

The pair-alignment statistic is particularly sensitive to this effect because it is not computed from individual directions alone, but from pair-separation directions. Pair directions depend on the joint distribution of two observed points:

$$P_{\text{pair}}(x_1, x_2) = P_{\text{obs}}(x_1) P_{\text{obs}}(x_2) \quad (\text{C.17})$$

Substituting equation (C.16) into equation (C.17), we obtain

$$P_{\text{pair}} \propto W(x_1) W(x_2) P_{\text{iso}}(r_1) P_{\text{iso}}(r_2) \quad (\text{C.18})$$

Thus, the geometry enters the statistic twice, through both members of each pair. This double appearance of the selection function explains why pair-alignment statistics can be strongly affected by observational geometry.

The axis that maximizes $A(\hat{n})$ may therefore correspond to the dominant angular structure of the selection function rather than an independent physical feature of the data. In this sense, the preferred direction can be induced by the observational footprint.

This is the geometric reason why an apparent cosmic axis may emerge even when the underlying distribution contains no intrinsic preferred direction.

C.4 Interpretation of Geometry-Preserving Nulls

The geometry-preserving null model is designed to preserve the observational selection structure while removing intrinsic object-level correlations.

In the spatial case, this means preserving the radial distribution and angular footprint of the observed catalog. Formally, the null realizations are drawn from the same observational measure,

$$d\mu_W(x) = W(x) d^3x \quad (\text{C.19})$$

A geometry-preserving null transformation T_{L2} satisfies

$$d\mu_{L2} = d\mu_W \quad (\text{C.20})$$

This means that the L2 null changes the realization of the catalog while preserving the observational measure itself.

Therefore,

$$E_{L2}[A(\hat{n})] \simeq E_W[A(\hat{n})] \quad (\text{C.21})$$

Equation (C.21) explains why L2 reproduces the observed anisotropy when the signal is generated by selection geometry. L2 is not merely another random baseline. It samples from the same observational geometry that generated the measured statistic.

In this sense, the agreement between A_{obs} and A_{L2} is not coincidental. It reflects the preservation of the same geometric constraint.

This is the central reason why L2 has interpretive priority over L1. If a signal disappears or becomes statistically typical under L2, then the signal is not uniquely attributable to intrinsic physical structure. It is compatible with the geometry of the observational domain.

Thus, L2 functions as a geometric explanatory baseline. It tests whether the observed anisotropy requires an intrinsic physical origin, or whether it can be generated by the same selection structure that defines the dataset.

This interpretation does not assume that intrinsic anisotropy is impossible. A geometry-preserving null may also absorb an intrinsic signal if that signal is perfectly aligned with, or statistically degenerate with, the observational geometry. In that case, the signal is not eliminated as a physical possibility; rather, its physical origin is not identifiable using the present statistic and null construction.

C.5 Failure of Naive Random Nulls

The naive isotropic null replaces the observational selection function with a uniform sampling domain. Formally, this corresponds to

$$W(x) \rightarrow W_{\text{iso}}(x) \quad (\text{C.22})$$

Under such a null, the expectation value becomes

$$E_{L1}[A(\hat{n})] \simeq \frac{1}{2} \quad (\text{C.23})$$

assuming an isotropic full-domain randomization.

However, this is not the correct expectation for a catalog drawn from a non-uniform observational domain. The observed catalog is not sampled from W_{iso} , but from W . Therefore, L1 and the observed data are generated under different geometric measures:

$$d\mu_{L1} \neq d\mu_W \quad (\text{C.24})$$

Consequently, the L1 null and the observed catalog are not generated under the same observational measure.

Therefore, the difference

$$A_{\text{obs}} - E_{L1}[A] \quad (\text{C.25})$$

does not isolate intrinsic physical anisotropy. It also contains the difference between the observational geometry and the naive isotropic domain.

This is why L1 can produce an apparently significant anisotropy even when the signal is geometry-induced. L1 removes the very geometric structure that shaped the observed catalog. The resulting comparison therefore confounds physical anisotropy with observational-domain mismatch.

By contrast, L2 preserves the observational measure:

$$d\mu_{L2} = d\mu_W \quad (\text{C.26})$$

Therefore, L1 and L2 are not interchangeable null baselines. They correspond to fundamentally different geometric assumptions. L1 removes the selection function, while L2 preserves it, leading to structurally distinct expectations for $A(\hat{n})$.

This explains why a signal can appear significant under L1 but become statistically ordinary under L2. The observed signal has not changed. The reference geometry has changed.

C.6 Implications for Non-Identifiability

The preceding sections establish that the observed anisotropy statistic can be written schematically as a function of both intrinsic structure and observational geometry:

$$A_{\text{obs}} = F(S_{\text{phys}}, G_{\text{obs}}) \quad (\text{C.27})$$

where S_{phys} denotes intrinsic physical structure and G_{obs} denotes observational geometry.

The central problem is that different pairs $(S_{\text{phys}}, G_{\text{obs}})$ may produce the same observed statistic:

$$F(S_{\text{phys}}, G_{\text{obs}}) = F(S'_{\text{phys}}, G'_{\text{obs}}) \quad (\text{C.28})$$

In particular, an observed signal may be compatible with

$$A_{\text{obs}} \simeq A_{\text{geom}} \quad (\text{C.29})$$

where A_{geom} is the anisotropy expected from the observational geometry alone.

When this occurs, the physical origin of the signal cannot be uniquely determined from the statistic itself. This is the sense in which the signal is non-identifiable.

The non-identifiability is not merely a matter of low statistical significance. It is a structural degeneracy between intrinsic anisotropy and geometry-induced anisotropy. Both can generate observationally similar outcomes.

This is why statistical detection and physical interpretation must be separated. A deviation from a naive null may establish that the catalog differs from an unconstrained random distribution, but it does not establish that the difference is physically intrinsic.

A physically interpretable anisotropy must therefore satisfy a stronger condition:

$$A_{\text{obs}} \neq A_{L2} \quad (\text{C.30})$$

Here, the inequality should be understood statistically:

the observed value must be distinguishable from the $L2$ distribution within the adopted significance criterion.

If instead

$$A_{\text{obs}} \simeq A_{L2} \quad (\text{C.31})$$

then the observed anisotropy is compatible with the geometry-preserving null and cannot be uniquely assigned to intrinsic cosmic structure within the present framework.

In this setting, the emergence of apparent anisotropy can follow from the geometry of the observational domain. Given a non-uniform selection function, directional dependence in the alignment statistic can become a natural consequence of the sampling configuration.

This establishes that apparent anisotropy need not imply new physics, but can arise as a structural property of the observational process itself.

C.7 Propagation to Inference-Level Quantities

The geometric origin of apparent anisotropy is not confined to spatial statistics. It can also propagate into inference-level quantities whenever those quantities are evaluated using direction-dependent partitions.

Consider an inference-level quantity Q_i assigned to each observed object. In the supernova application discussed in the main text, this role is played by an effective H_0^{proxy} quantity, denoted by $H_{0,i}^{\text{proxy}}$. A directional contrast can be defined by splitting the sample relative to a trial axis $\hat{n}$.

Here, the subscript proxy is used only as a descriptive label indicating that the quantity is an effective estimator, not a directly measured global Hubble constant.

For example,

$$\Delta H_0^{\text{proxy}}(\hat{n}) = \langle H_0^{\text{proxy}} \rangle_{\parallel}(\hat{n}) - \langle H_0^{\text{proxy}} \rangle_{\perp}(\hat{n}) \quad (\text{C.32})$$

where the parallel and perpendicular subsets are defined by the angular alignment between each object and the trial axis $\hat{n}$.

The important point is that this partition is itself geometry-dependent. The subsets are not formed independently of the survey footprint; they are selected from the same non-uniform angular distribution $W_{\text{ang}}(\hat{x})$.

Thus, the expectation value of the directional contrast is not determined only by the physical values $H_{0,i}^{\text{proxy}}$. It also depends on the observational geometry:

$$E[\Delta H_0^{\text{proxy}}(\hat{n})] = G(H_0^{\text{proxy}}, W_{\text{ang}}, W_{\text{rad}}, \hat{n}) \quad (\text{C.33})$$

If the angular footprint is non-uniform, then the parallel and perpendicular subsets may differ in sky coverage, redshift distribution, sampling density, or duplicate structure even when no intrinsic directional physics is present.

A geometry-preserving inference-level null therefore keeps the spatial configuration fixed and randomly permutes the scalar values across the observed positions. In schematic form,

$$(x_i, H_{0,i}^{\text{proxy}}) \rightarrow (x_i, H_{0,\pi(i)}^{\text{proxy}}) \quad (\text{C.34})$$

where π is a random permutation.

This preserves the observational geometry while removing any intrinsic association between direction and the inferred scalar quantity. If the observed contrast remains statistically consistent with this null, then the directional inference is not physically identifiable.

Therefore, consistency with geometry-preserving null models at the level of $\Delta H_0^{\text{proxy}}$ indicates that the apparent directional variation cannot be independently attributed to intrinsic physical anisotropy.

This demonstrates that non-identifiability is not confined to spatial statistics. It is a structural property of the observational process that can persist across both spatial and inference-level analyses.

In this sense, the AIP framework applies not only to anisotropy statistics such as $A(\hat{n})$, but also to derived cosmological quantities whenever their directional interpretation depends on the same observational geometry.

C.8 Summary of Appendix C

Appendix C establishes the geometric basis of the non-identifiability result.

The central point is that a non-uniform observational domain changes the expectation value of directional statistics. Even if the underlying distribution is isotropic, the observed distribution inherits the angular and radial structure of the selection function.

Therefore, apparent anisotropy can arise from the observational process itself.

The geometry-preserving null model reproduces this effect because it preserves the observational measure. The naive null fails because it replaces the observed selection geometry with an unconstrained isotropic domain.

Consequently, a signal that is significant under L1 but statistically consistent under L2 cannot be uniquely interpreted as intrinsic physical anisotropy.

The same logic extends to inference-level quantities such as direction-dependent H_0^{proxy} estimates, because these quantities are evaluated through directional partitions defined on the same non-uniform observational geometry.

Thus, the core conclusion of this appendix is:

$$A_{\text{obs}} \simeq A_{L2} \tag{C.35}$$

This relation implies non-identifiability of the physical origin within the geometry-preserving framework.

This does not prove that intrinsic anisotropy is absent. Rather, it shows that intrinsic anisotropy is not required to explain the observed statistic within the geometry-preserving framework.